

The Circulatory System

Why is a Transport System needed?

Unicellular organisms do not require a transport system as they are one cell thick, allowing diffusion to help them absorb substances and excreted waste products.

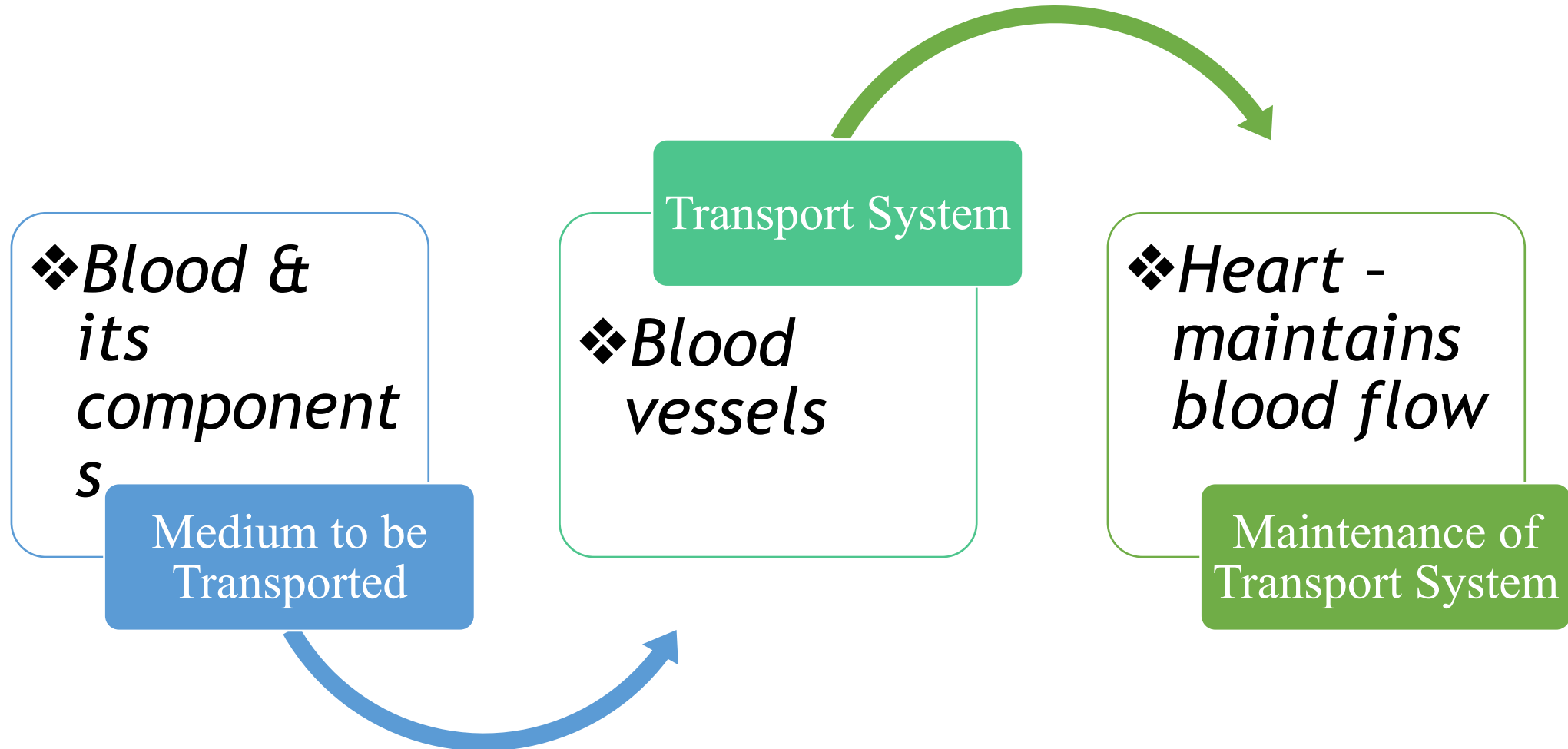
Multicellular organisms on the other hand have a *small surface area to volume ratio* and are therefore unable to use diffusion as their main mode of transport.

Thus, they develop a transport system to get all the substances that their cells require and remove their waste products of metabolism.

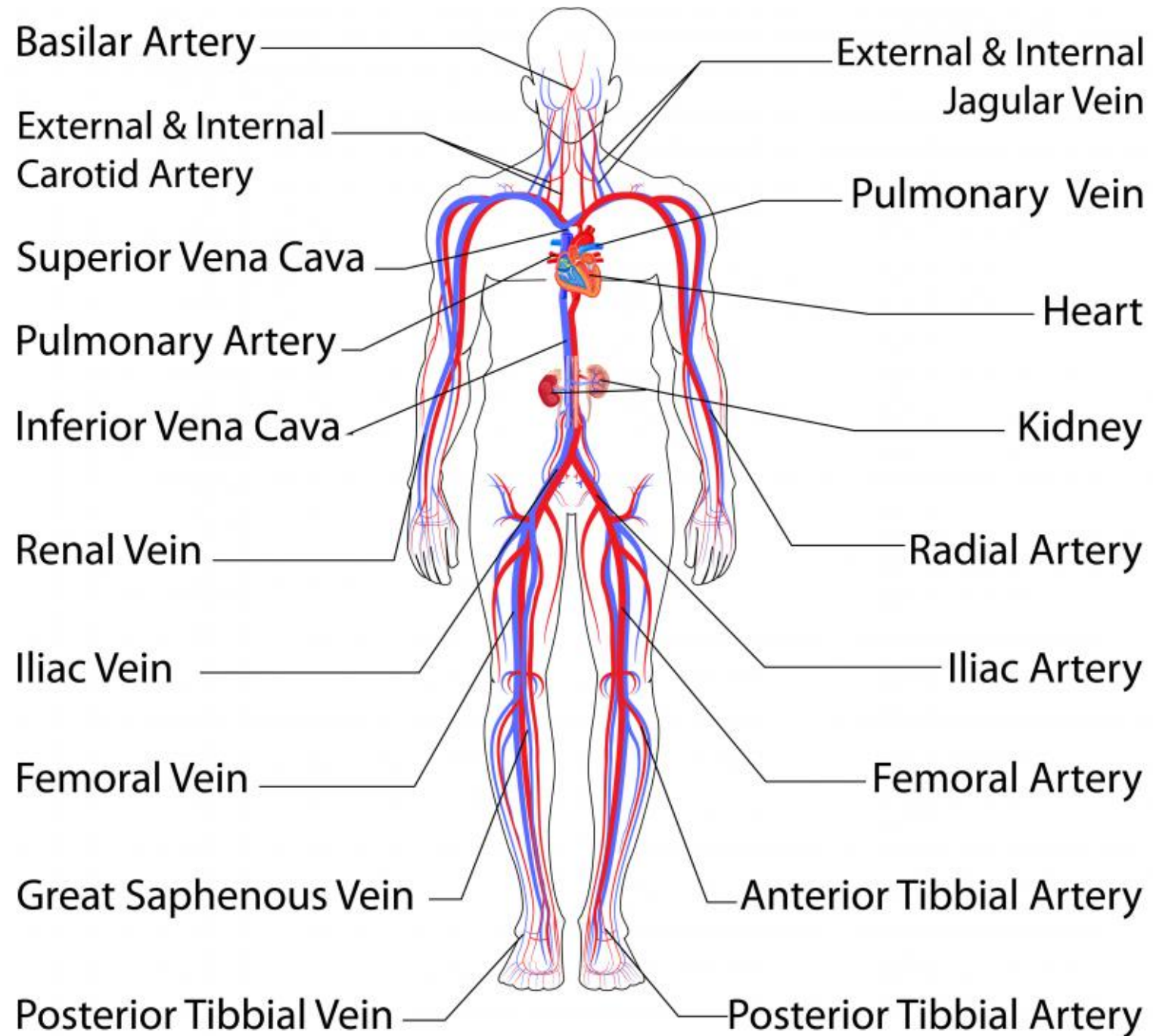
Substances to be Transported

- Glucose, amino acids, fats
- Oxygen
- Mineral salts
- Hormones
- Antibodies
- Water
- Carbon dioxide
- Nitrogenous waste

What constitutes to a Transport System?



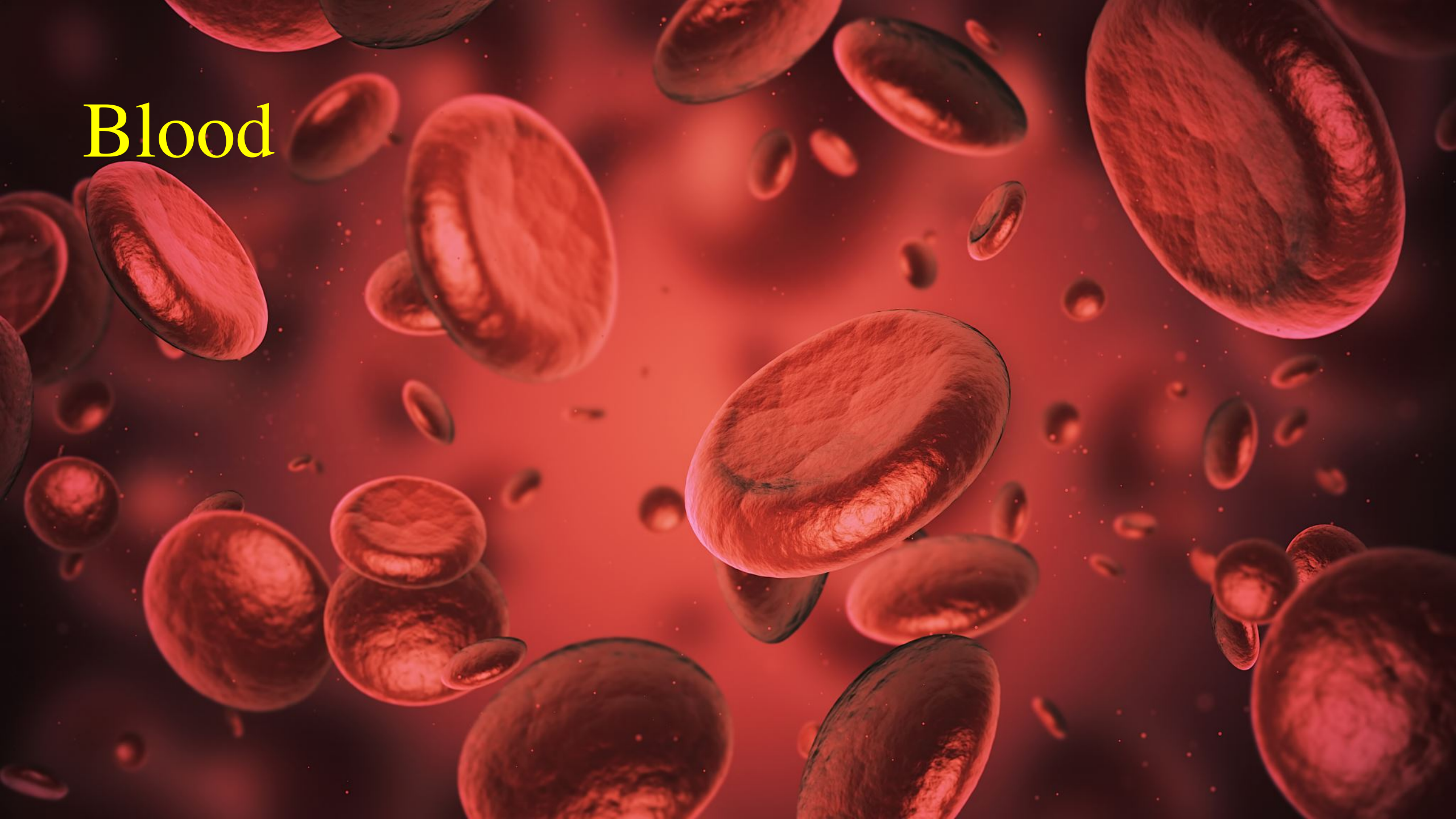
The Circulatory System



Functions of the Circulatory System

- ***Transportation of Gases***
- ***Nutrient Transport***
- ***Waste Removal***
- ***Hormone Transport***
- ***Thermoregulation***

Blood



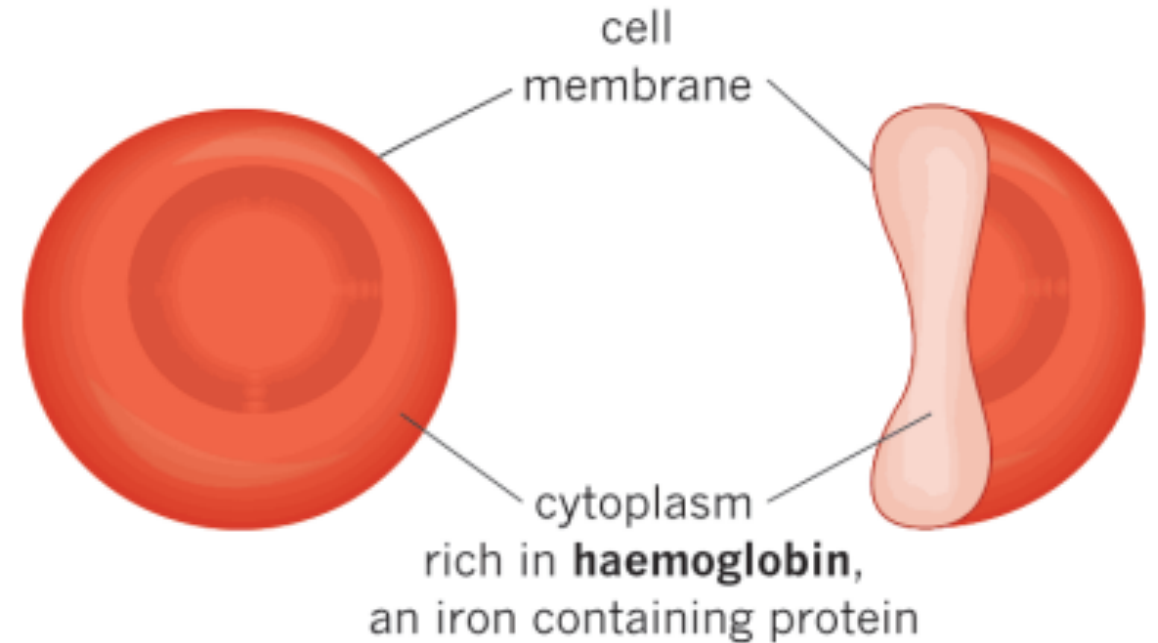
Blood

Blood consists of plasma which houses all the substances to be transport made up of 90% water and 10% dissolved substances. It also contains a few essential cells and proteins;

- ✓ Red blood cells
- ✓ White blood cells
- ✓ Platelets
- ✓ Plasma proteins; fibrinogen, prothrombin, albumin, antibodies
- ✓ Mineral salts
- ✓ Hormones

Red Blood Cells / Erythrocytes

- Biconcave discs with a center and a relative larger surface area to volume ratio for gasses to diffuses in & out.
- No nucleus and lives for 3-4 months.
- Contains the red pigment haemoglobin.
- Slightly elastic, allowing them to squeeze through the narrowest of capillaries.



Red Blood Cells

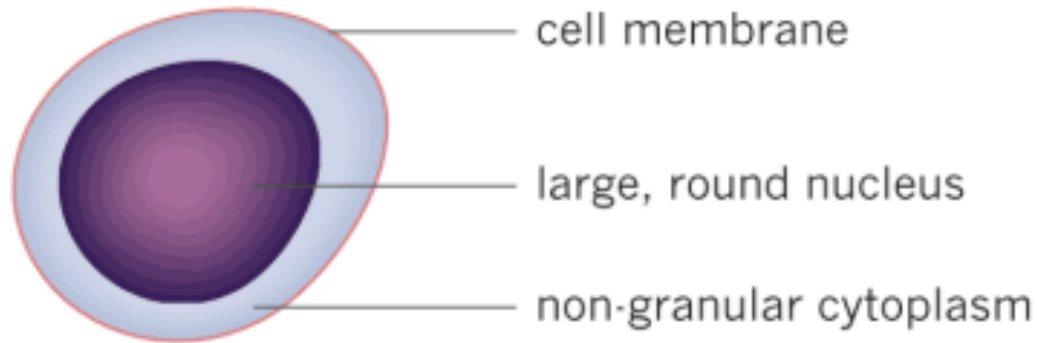
Formation of Cells:

- Formed in the red bone marrow found in flat bones and in the end of long bones.
- Broken down mainly in the liver and spleen.

Functions:

- Transport oxygen as oxyhaemoglobin from the lungs to body cells.
- Transport small amounts of carbon dioxide from the body cells to the lungs.

White Blood Cells - Lymphocytes



✓ Makes up 25% of white blood cells.

✓ Have a rounded shape.

✓ Have a large rounded nucleus that controls the production of antibodies.

✓ Have only a small amount of cytoplasm.

White Blood Cells - Lymphocytes

Formation:

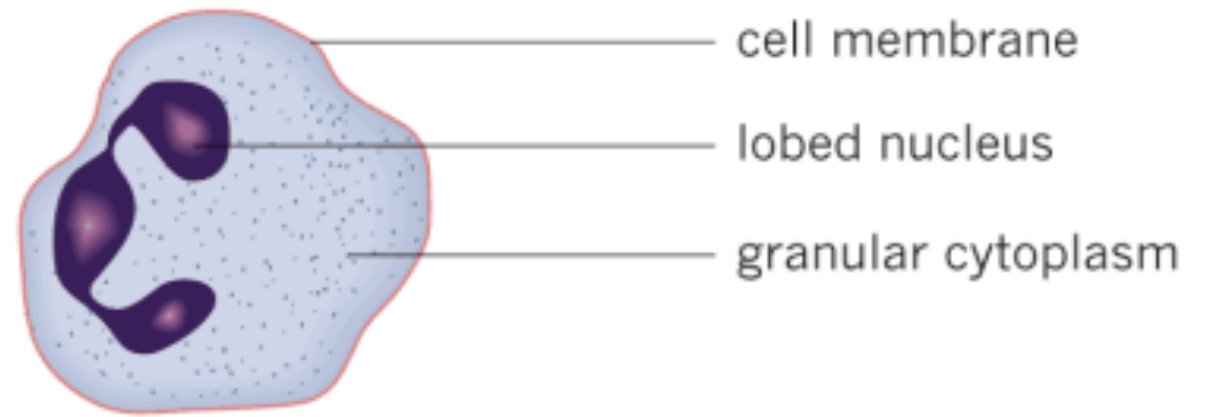
- Developed from the cells in the red bone marrow and mature in other organs e.g. lymph nodes, spleen and thymus gland

Function:

- ✓ Produce antibodies to destroy disease causing pathogens.
- ✓ Produce antitoxins to neutralize toxins produced by pathogens.

White Blood Cells - Phagocytes

- 75% of WBCs
- Have a variable shape.
- Have a lobed nucleus.
- Move by pseudopodia; can move out of capillaries through their walls and engulf pathogens using pseudopodia.



White Blood Cells - Phagocytes

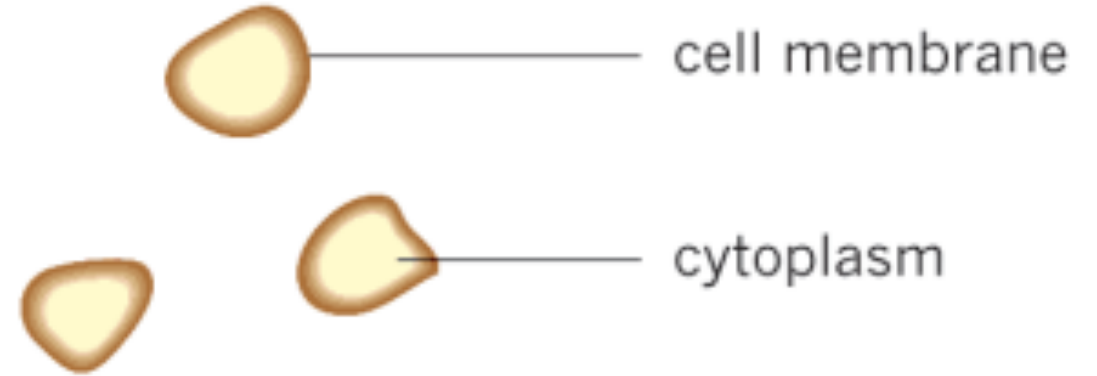
Formation:

- ❑ Formed in the red bone marrow.

Function:

- ❖ Engulf and destroy pathogens.
- ❖ Engulf pathogens destroyed by antibodies.

Platelets



- Are composed of cell fragments.
- They have no nucleus and live for about 10 days.

Function:

- Helps blood to clot at a cut or wound.

Formation:

- Formed from cells in the red bone marrow.

What are 4 Functions of Blood?

1)

2)

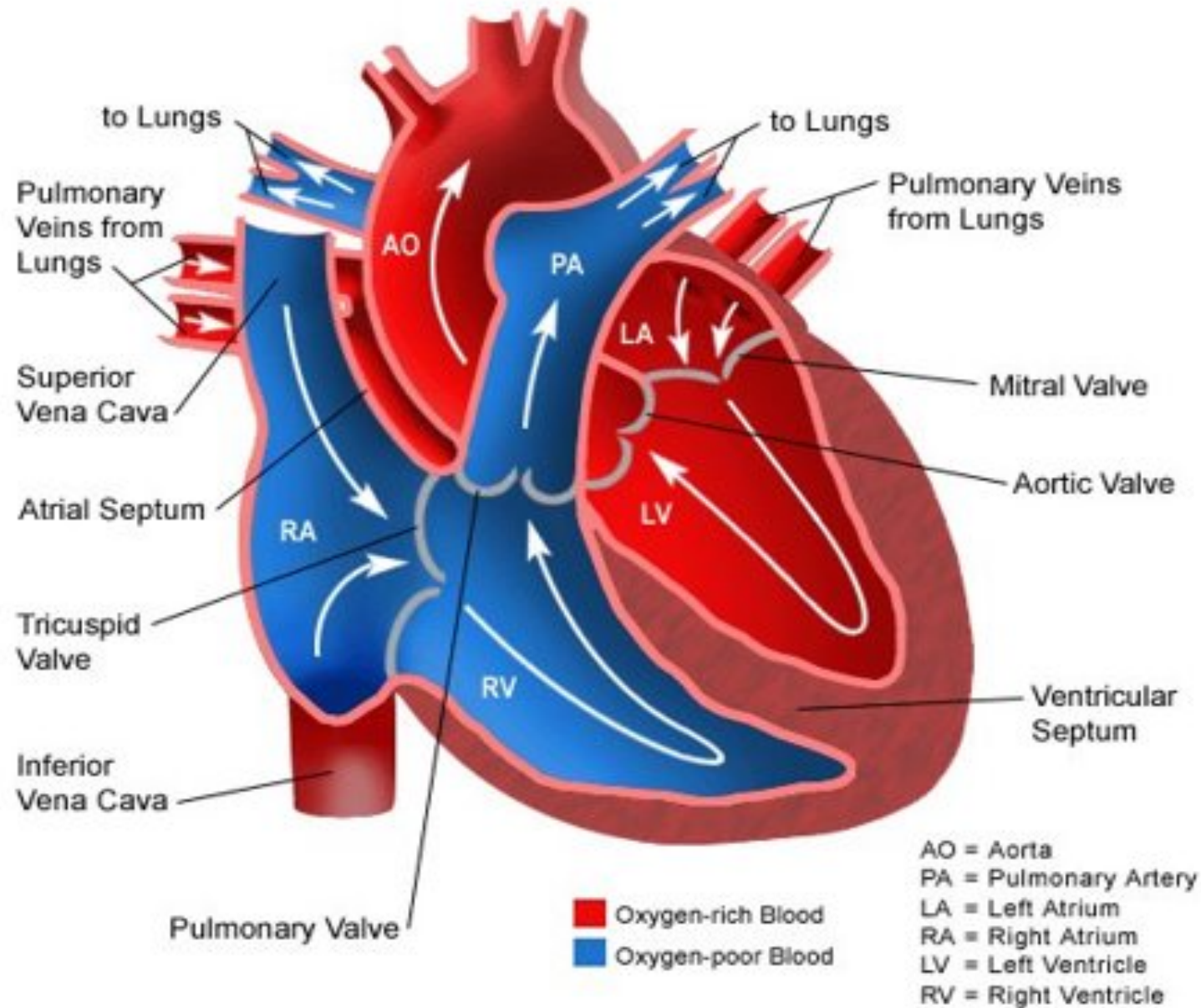
3)

4) .

The Heart



Normal Heart



The Heart

The heart is made up of 4 chambers; 2 upper chambers called atria and 2 lower chambers called ventricles.

The **right atrium and right ventricle** is separated by the *tricuspid valve* and contains deoxygenated blood.

The **left atrium and ventricle** is separated by the *bicuspid valve* and contains oxygenated blood.

The right side of the heart is separated from the left side of the heart by a *septum*.

The Heart

Right Side

- Deoxygenated blood
- Ventricle has a thinner muscular wall
- Right atrium and ventricle separated by the tricuspid valve
- Blood is pumped to the lungs

Left Side

- Oxygenated blood
 - Ventricle has a thicker muscular wall
 - Left atrium and ventricle separated by the bicuspid/mitral valve
 - Blood is pumped to the rest of the body
-

Action of the Heart

The atria begins contracting as a result of a signal sent from the [Sino atrial node](#).

The SA node is a specialized tissue that lies in the [right atrium](#) that tells both atria to contract.

When both atria contract, this is called [atrial systole](#) and at this time the ventricles are relaxed to receive the blood, referred to as [ventricular diastole](#).

Atrial Systole

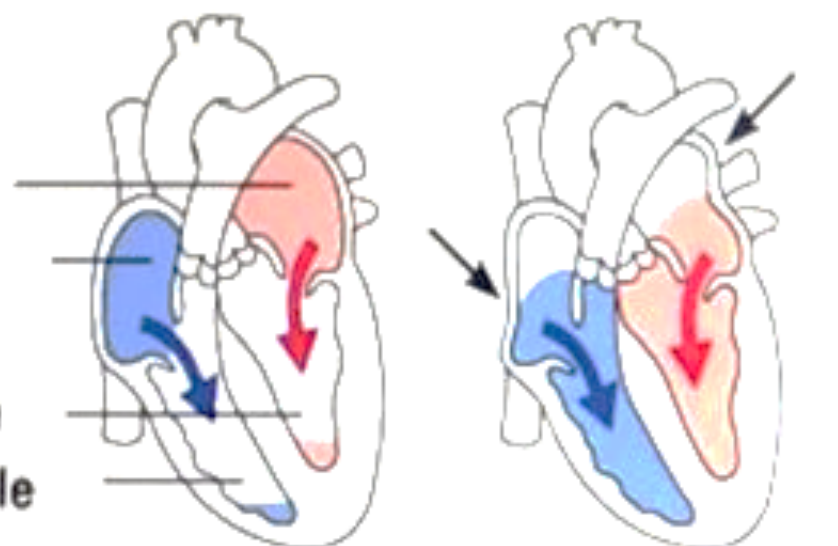


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Atrial
systole

Left atrium
Right atrium
Left ventricle
Right ventricle

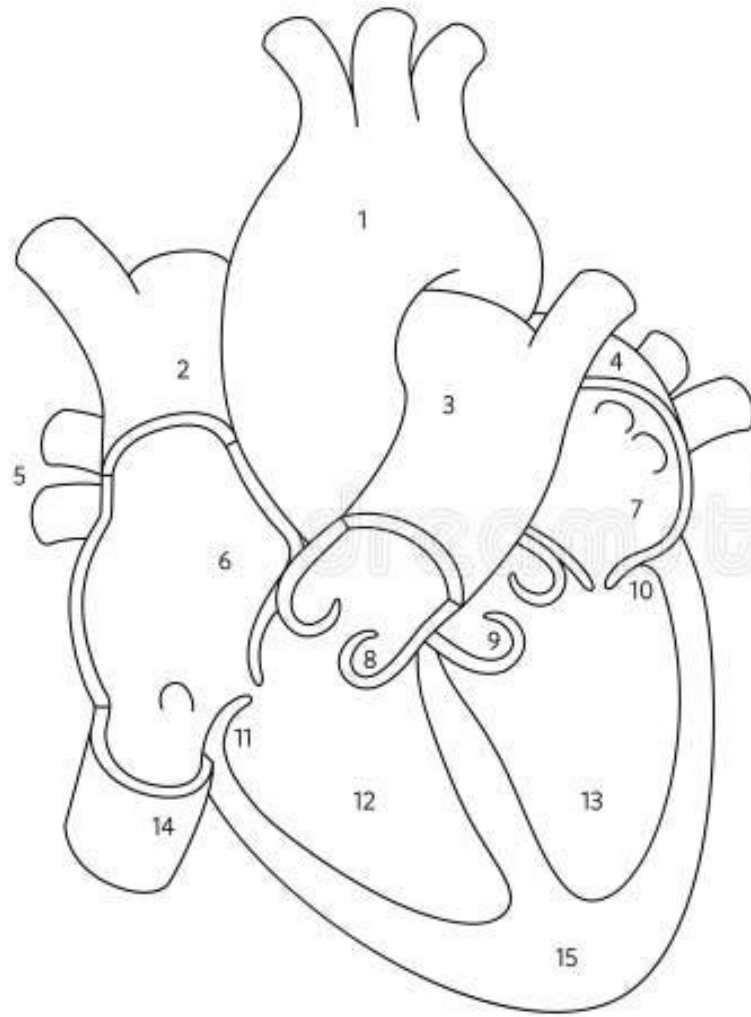


Ventricular
filling

Atrial
contraction

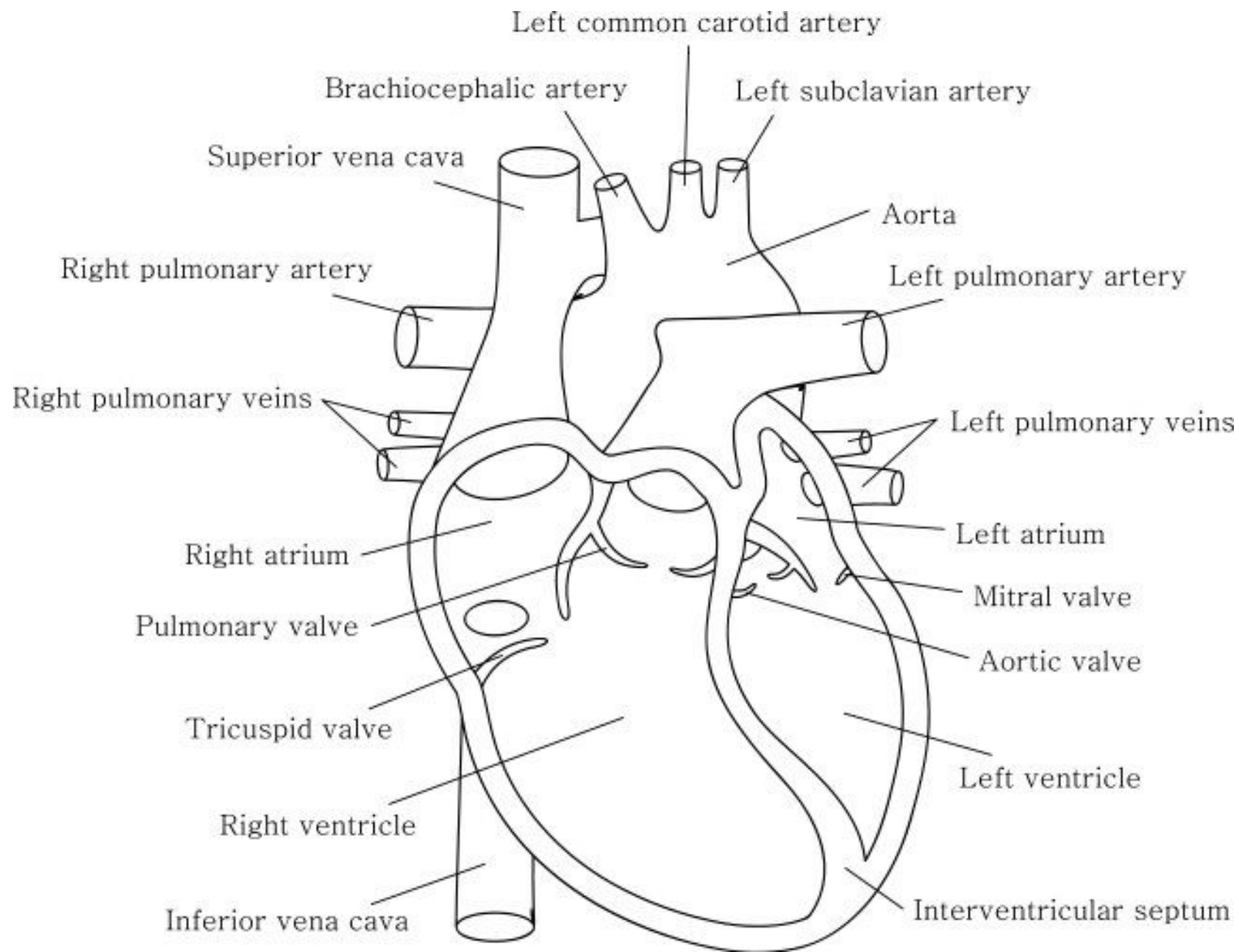
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(Ventricular filling)
mid-to-late diastole



Heart Anatomy

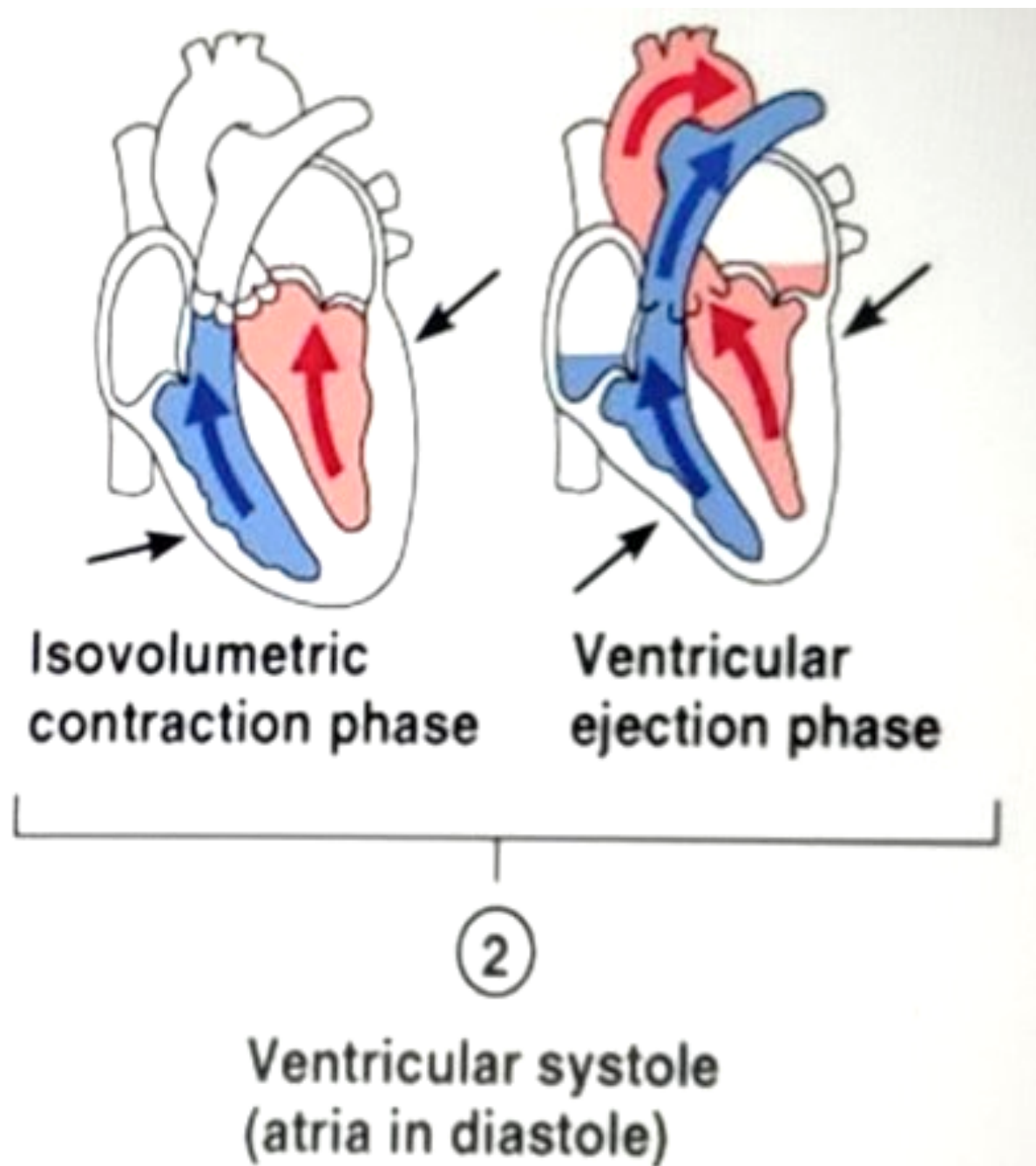
1. aorta
2. superior vena cava
3. pulmonary artery
4. pulmonary vein
5. right pulmonary veins
6. right atrium
7. left atrium
8. pulmonary valve
9. aortic valve
10. mitral valve
11. tricuspid valve
12. right ventricle
13. left ventricle
14. inferior vena cava
15. septum



Action of the Heart

The electrical impulse given by the SA node goes to the **AV (atrioventricular) node** which tells the ventricles to contract (*ventricular systole*) and the atria relaxes (*atria diastole*) to receive blood again.

- ❖ When the ventricles contract, blood from the right ventricle goes through the pulmonary artery to the lungs to undergo gaseous exchange.
- ❖ While the blood in the left ventricle goes through the aorta to the rest of the body.



➤ Ventricular systole

➤ Atria diastole

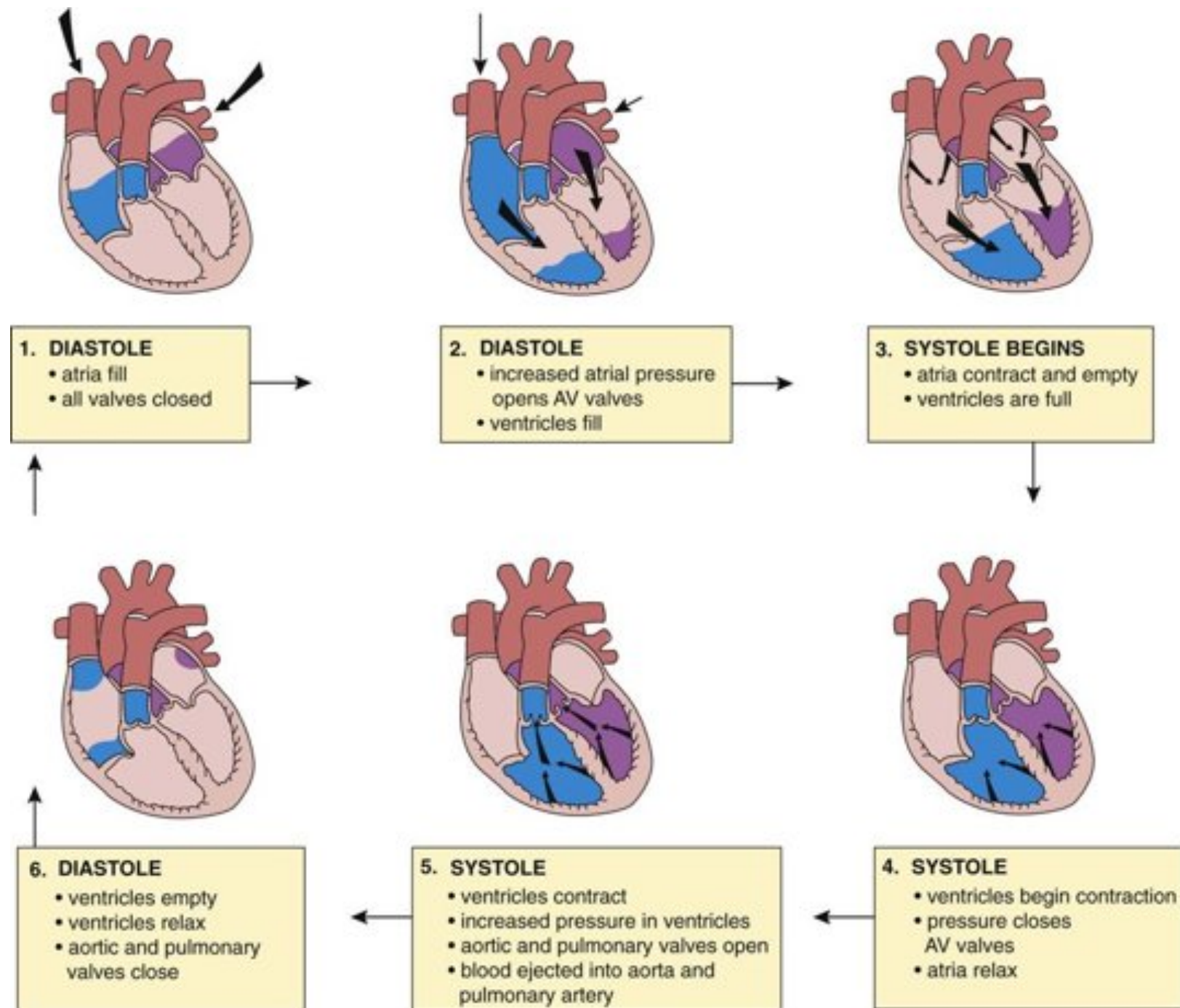
➤ Atria filling

Cardiac Cycle:

✓ Diastole

✓ Atrial Systole

✓ Ventricular Systole



The Heartbeat

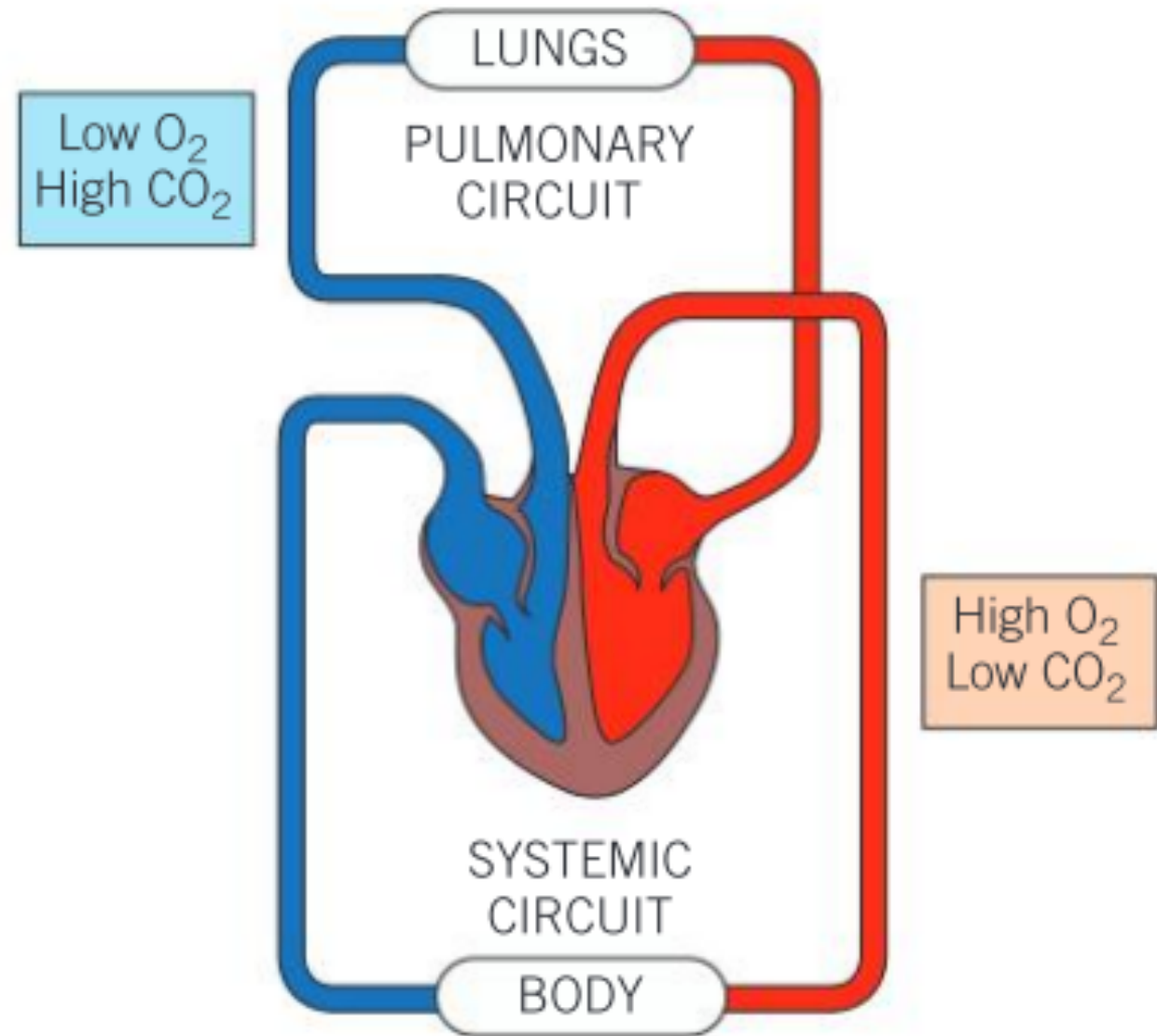
The lub-dub sound of the heartbeat is heard due to the closure of the heart valves:

- ❖ Lub – closure of the bicuspid & tricuspid valves (louder sound)
- ❖ Dub – closure of the semi-lunar valves (softer sound)

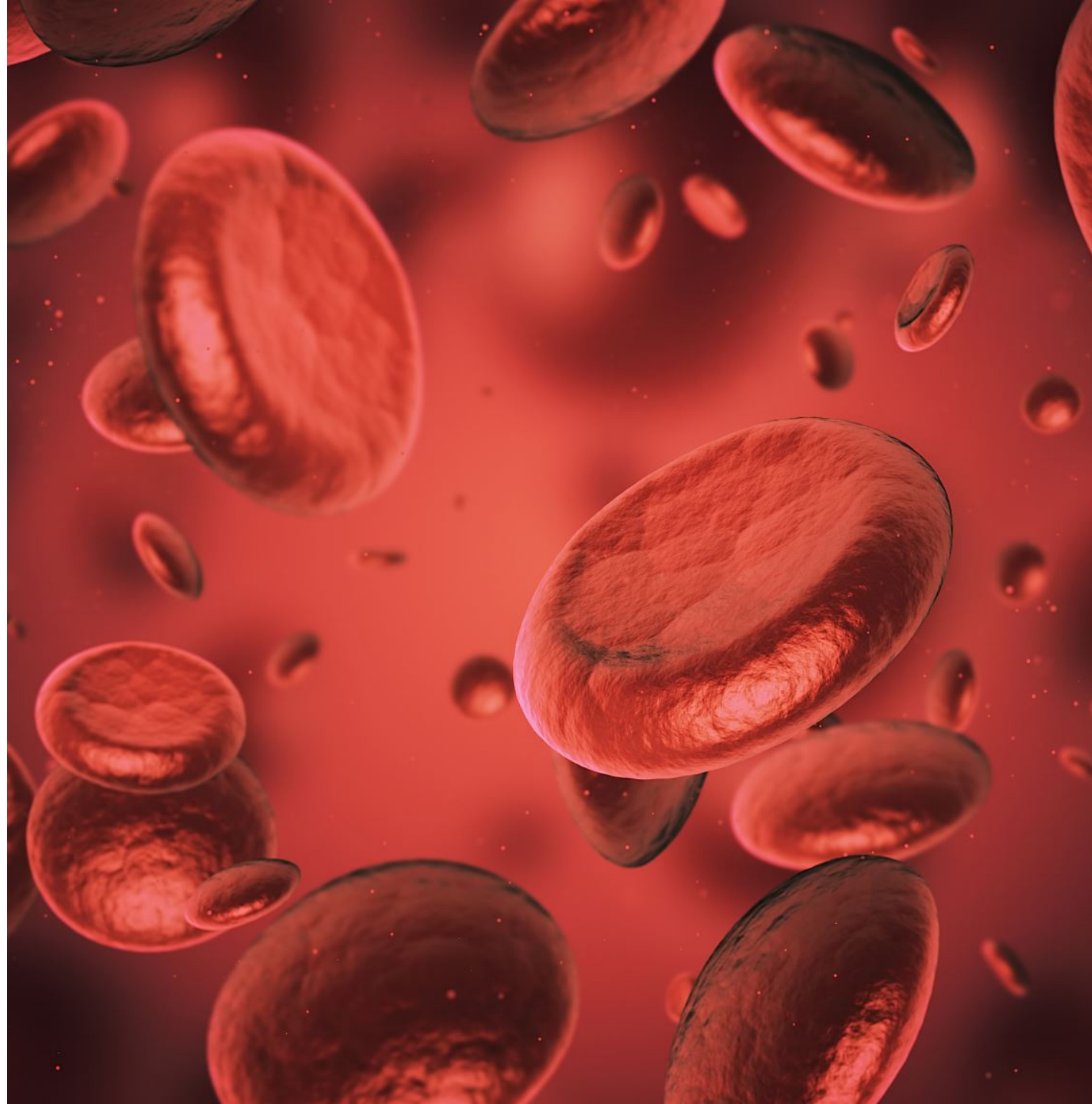
Circulation

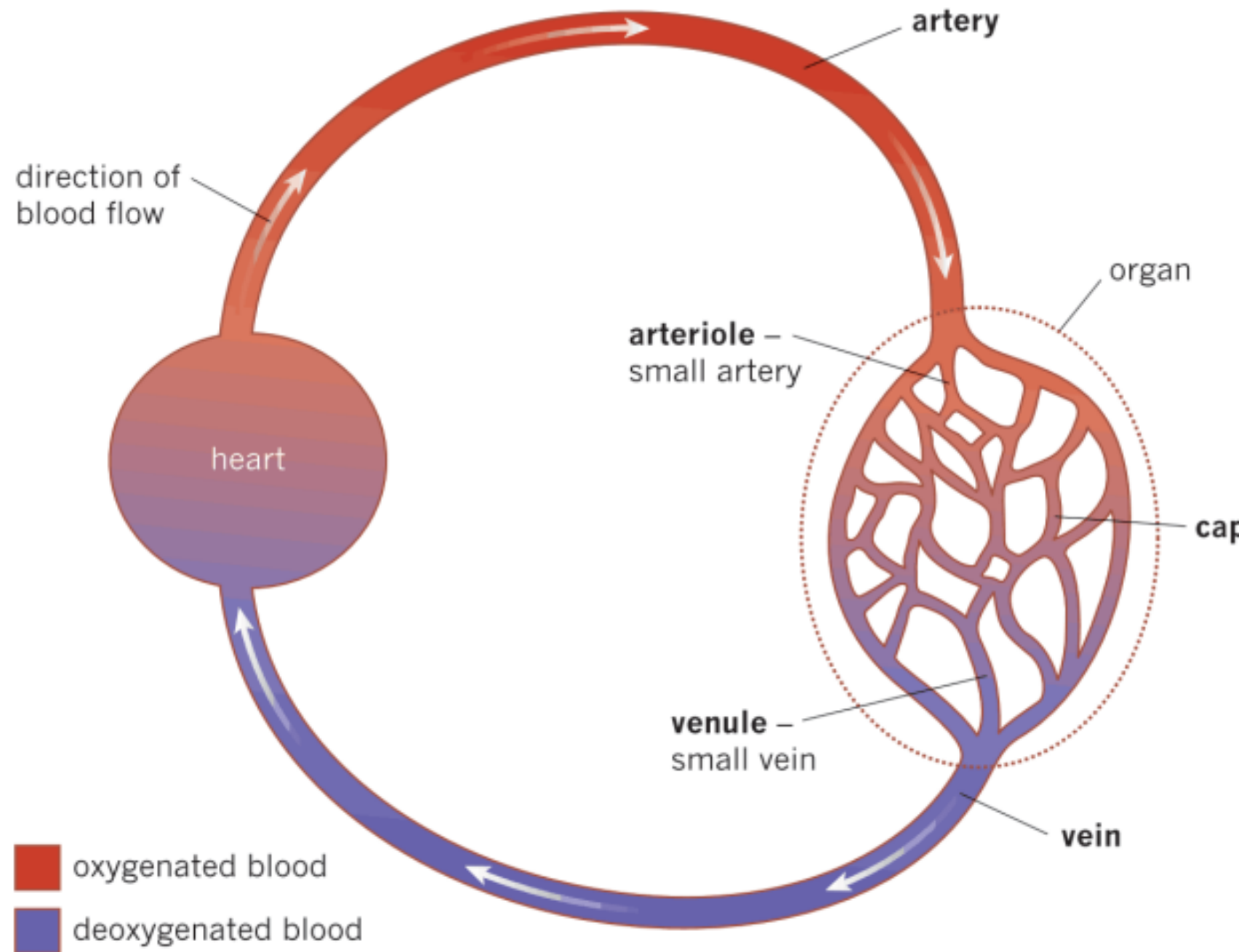
The human body has a double circulation:

- ✓ Pulmonary Circulation
- ✓ Systemic Circulation



The Blood Vessels





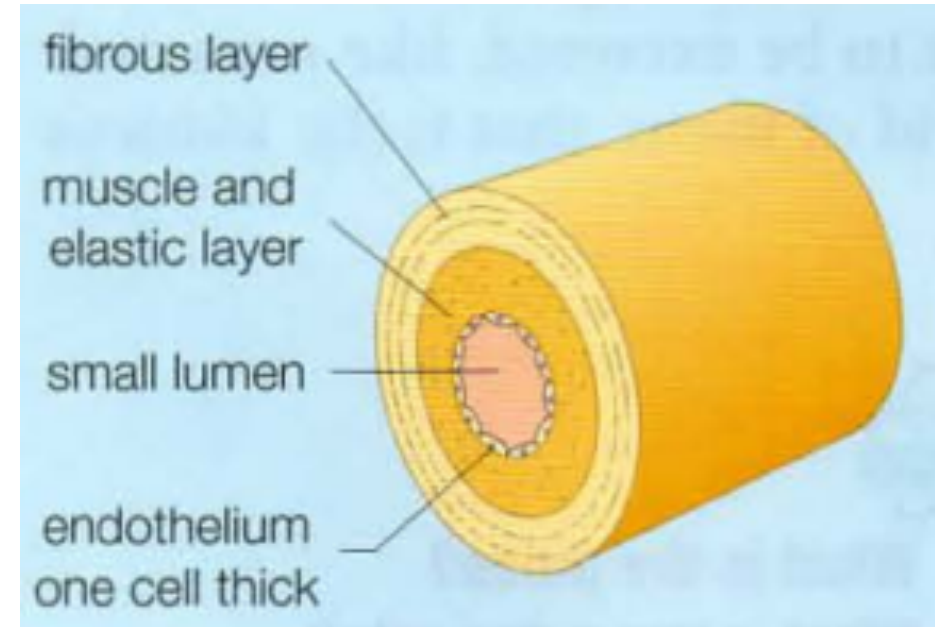
The Blood Vessels

The 3 main blood vessels:

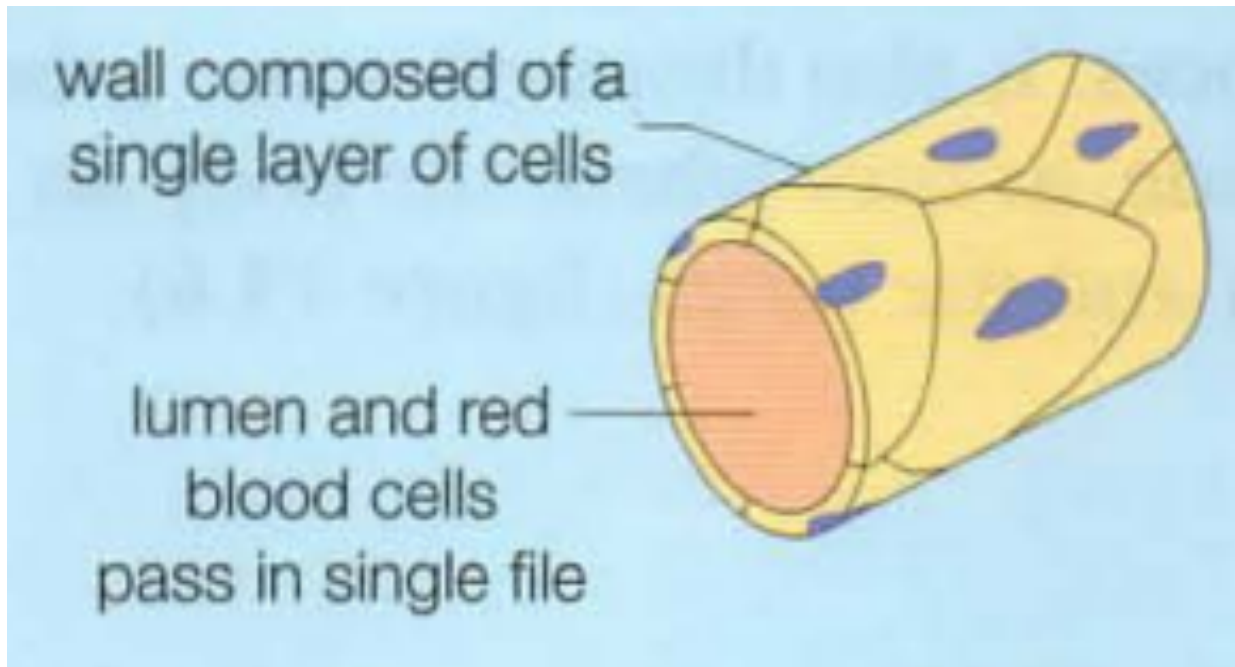
- I. Arteries
- II. Capillaries
- III. Veins

Arteries

- ❖ Arteries carry blood under high pressure and therefore have thick muscular & elastic walls to withstand the pressure.
- ❖ It carries blood away from the heart and flows in pulses.
- ❖ There are no valves present & they lie deep within the body.



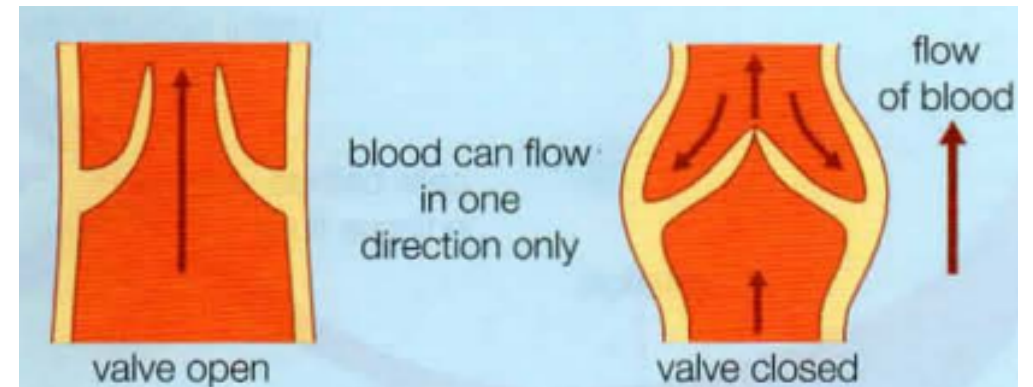
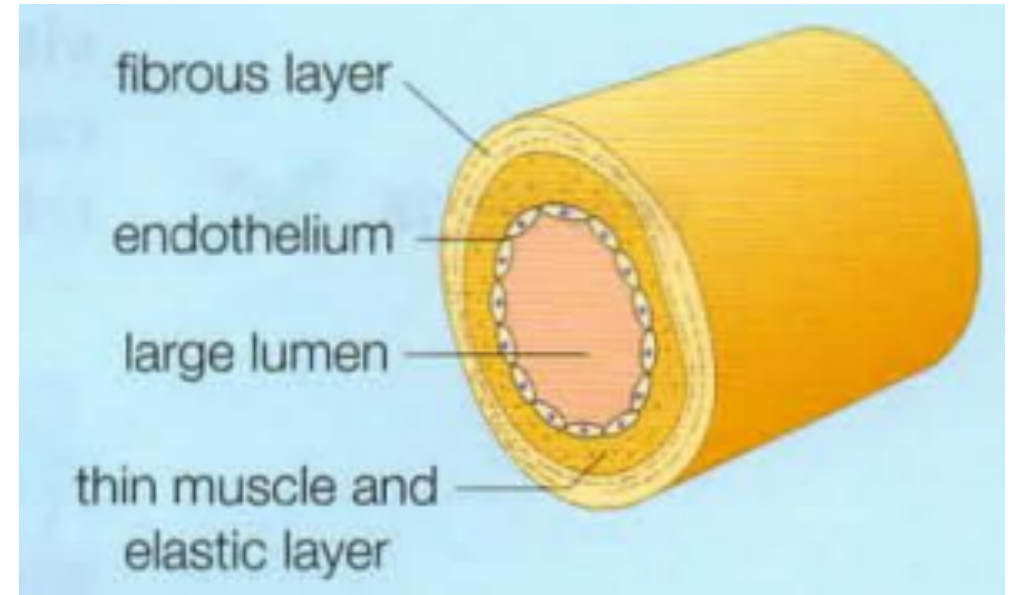
Capillaries



- ❖ Capillaries are one cell thick, allowing for easy diffusion.
- ❖ It runs through the tissues and there are no valves present.
- ❖ Blood flow is slow and smooth.

Veins

- ❖ Veins carry deoxygenated blood under low pressure to the heart. (smooth & slow flow)
- ❖ They have valves to prevent the backflow of blood.
- ❖ They lie close to the body surface.



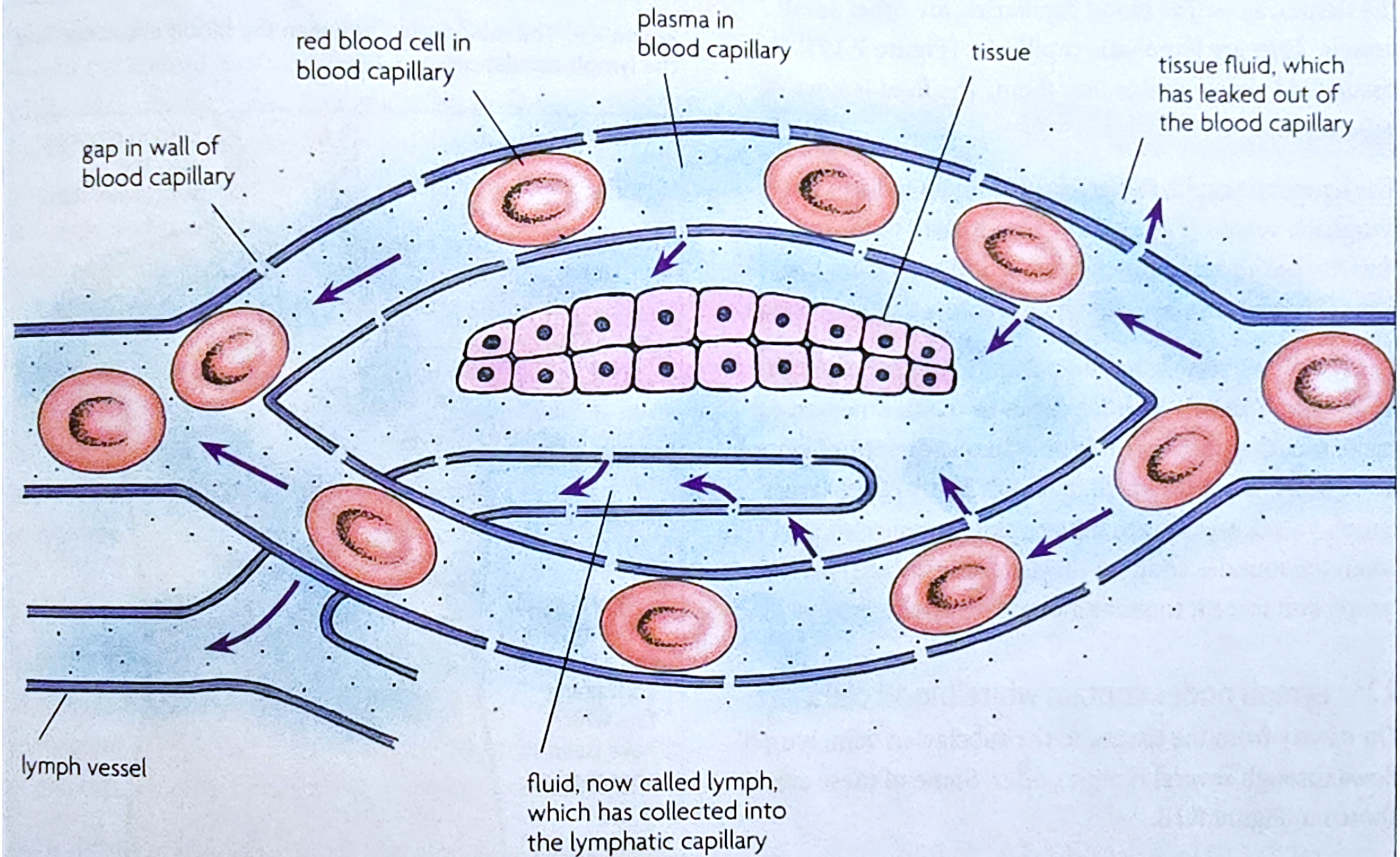
Blood Supply of Organs

Organ	Blood Supply	
	Artery	Vein
Brain	Common Carotid artery	Internal Jugular vein
Liver	Hepatic artery	Hepatic vein
Intestines	Mesenteric artery	Portal Hepatic vein
Kidney	Renal artery	Renal vein
Lungs	Pulmonary vein (carried oxygenated blood from the lungs to the heart)	Pulmonary artery (carries deoxygenated blood from the heart to the lungs)

Lymph & Tissue Fluid

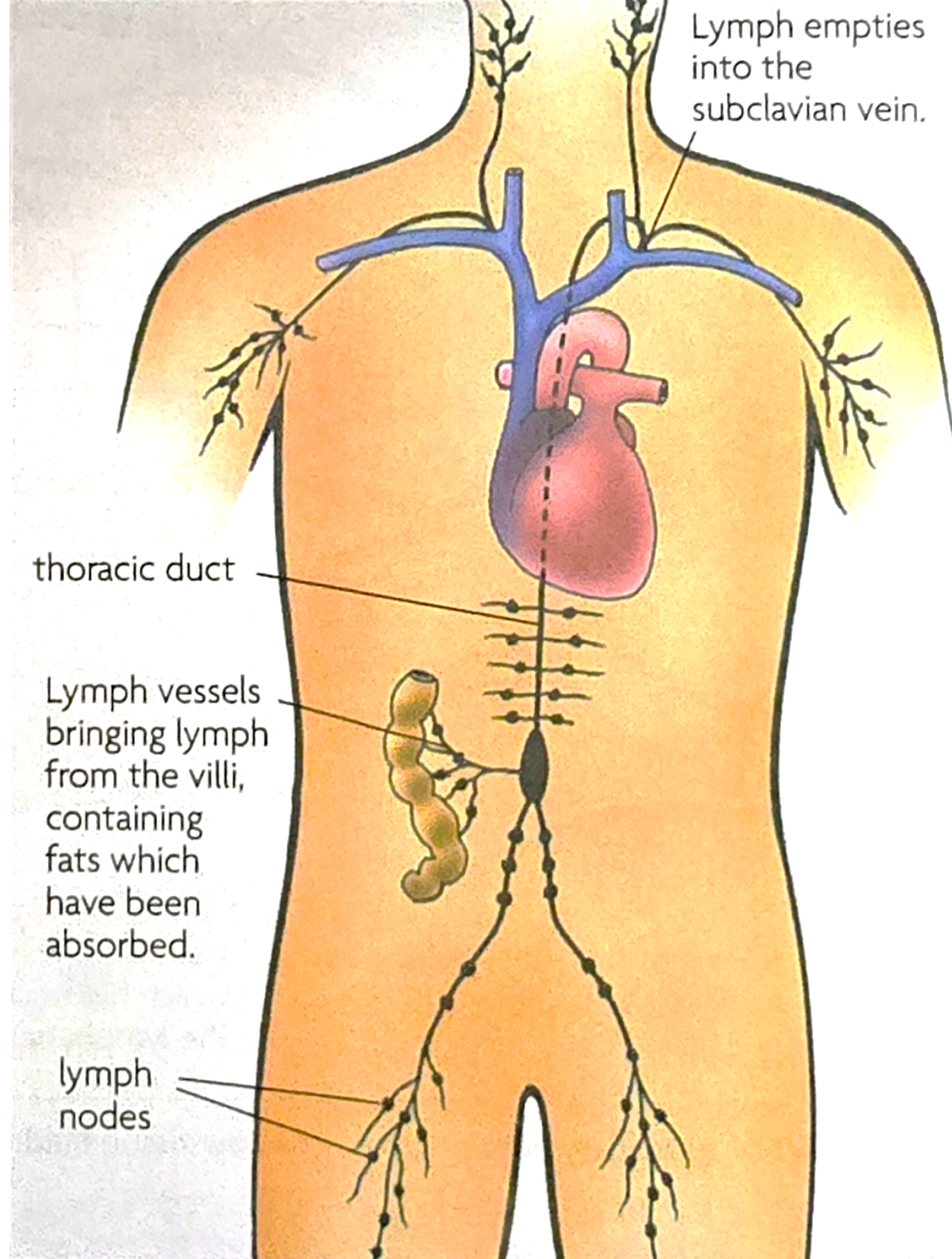
Tissue Fluid

- Capillaries leak plasma as there are small gaps between them.
- White blood cells and plasma are able to fit through these gaps.
- The fluid formed in this manner is called *tissue fluid* and it surrounds all cells in the body.
- This is because its function is to supply the cells with requirements they need.



Tissue fluid & Lymph

- ✓ Tissue fluid helps to maintain a constant environment for the body cells i.e. constant temperature and osmotic concentration.
- ✓ The plasma and white blood cells must eventually return to the blood through small vessels along side blood capillaries and in tissues called *lymphatic capillaries*.
- ✓ The fluid in these lymphatic capillaries is now referred to as *lymph*. Here the lymph is transported to the subclavian veins where it enters the blood again.



Lymph empties
into the
subclavian vein.

thoracic duct

Lymph vessels
bringing lymph
from the villi,
containing
fats which
have been
absorbed.

lymph
nodes

Lymph

- ◆ The lymph system does not have a pump but they do have valves in them to ensure movement in one direction.
- ◆ The larger lymph vessels lie very closely or within muscles so that when they contract, the lymph is forced to move along its vessels.
- ◆ Lymph nodes contain a large number of white cells allowing most bacteria and toxins to be destroyed/killed here.

Blood Groups



Blood Groups

The ABO system divides the blood into 4 groups:

- ✓ group A
- ✓ group B
- ✓ group AB
- ✓ group O

These groups can have the presence of 2 antigens and 2 antibodies accompanying them in the blood.

The antigen and antibody must be different to each other. If they were identical, it would cause agglutination or clumping of red blood cells.

Blood Groups

Blood Group	Antigen present on the surface of RBC	Antibody present in the plasma
Group A	A	Anti – B
Group B	B	Anti - A
Group AB	Both A & B	No antibodies
Group O	No antigens	Anti – A & anti – B

Rhesus Blood Group System

The Rhesus or Rh system further divides the blood into 2 groups:



Rh positive



Rh negative

This is determined by the presence or absence of the rhesus factor known as Rh factor.

Similarly, a Rh-positive person can receive blood from another Rh positive or Rh-negative person, but a Rh-negative person can only receive blood from another Rh-negative person.

Blood Transfusions

		Donor's blood type							
		O-	O+	B-	B+	A-	A+	AB-	AB+
Recipient's blood type	AB+	✓	✓	✓	✓	✓	✓	✓	✓
	AB-	✓		✓		✓		✓	
	A+	✓	✓			✓	✓		
	A-	✓				✓			
	B+	✓	✓	✓	✓				
	B-	✓		✓					
	O+	✓	✓						
	O-	✓							

Coagulation Cascade

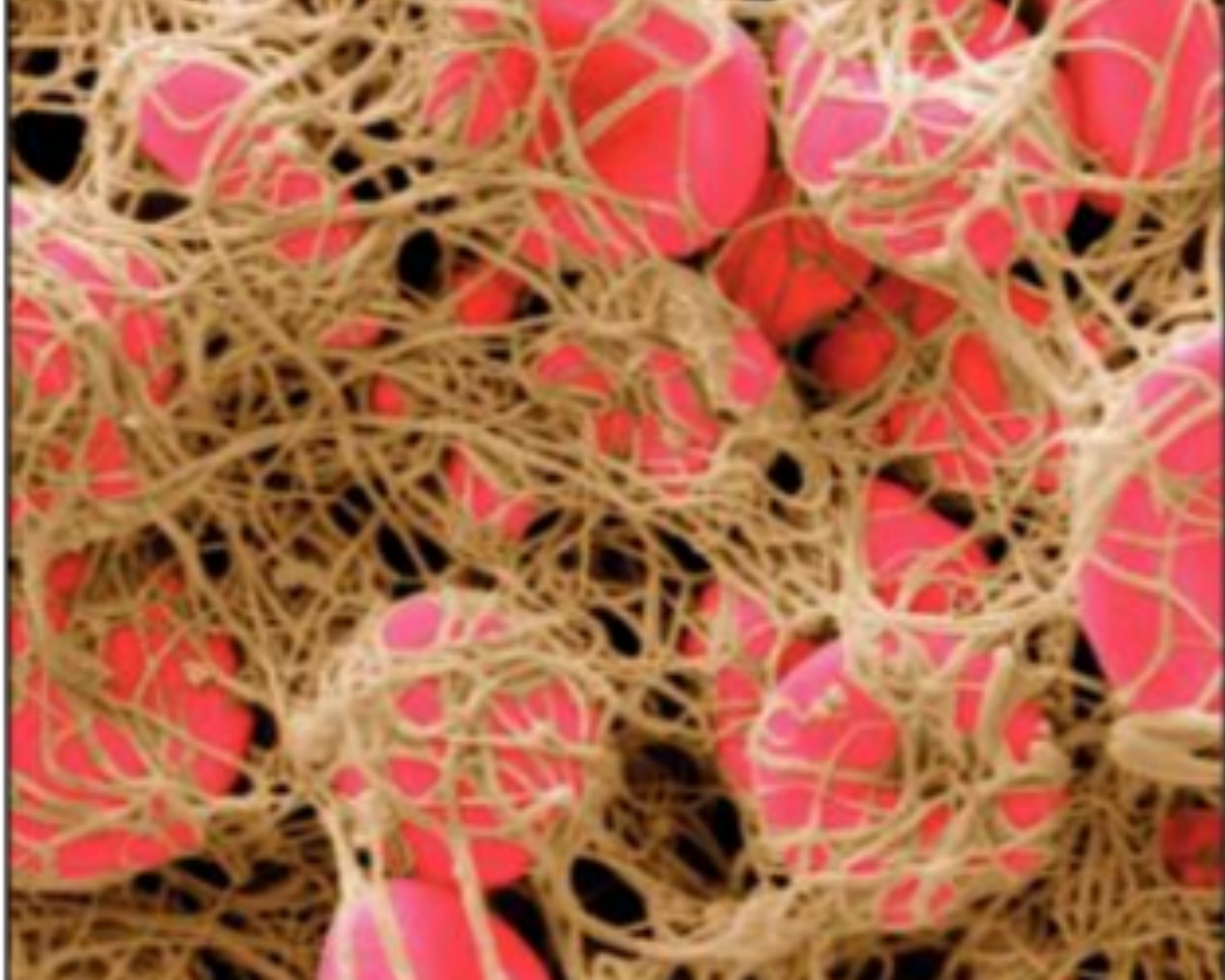
Importance of Blood Clotting

When a tissue is wounded, blood flows from it and coagulates to form a clot.

This prevents further blood loss and the entry of pathogenic microorganisms.

The clotting process depends on at least 12 clotting factors working in harmony with each other.

How does
blood clot?



The Wound

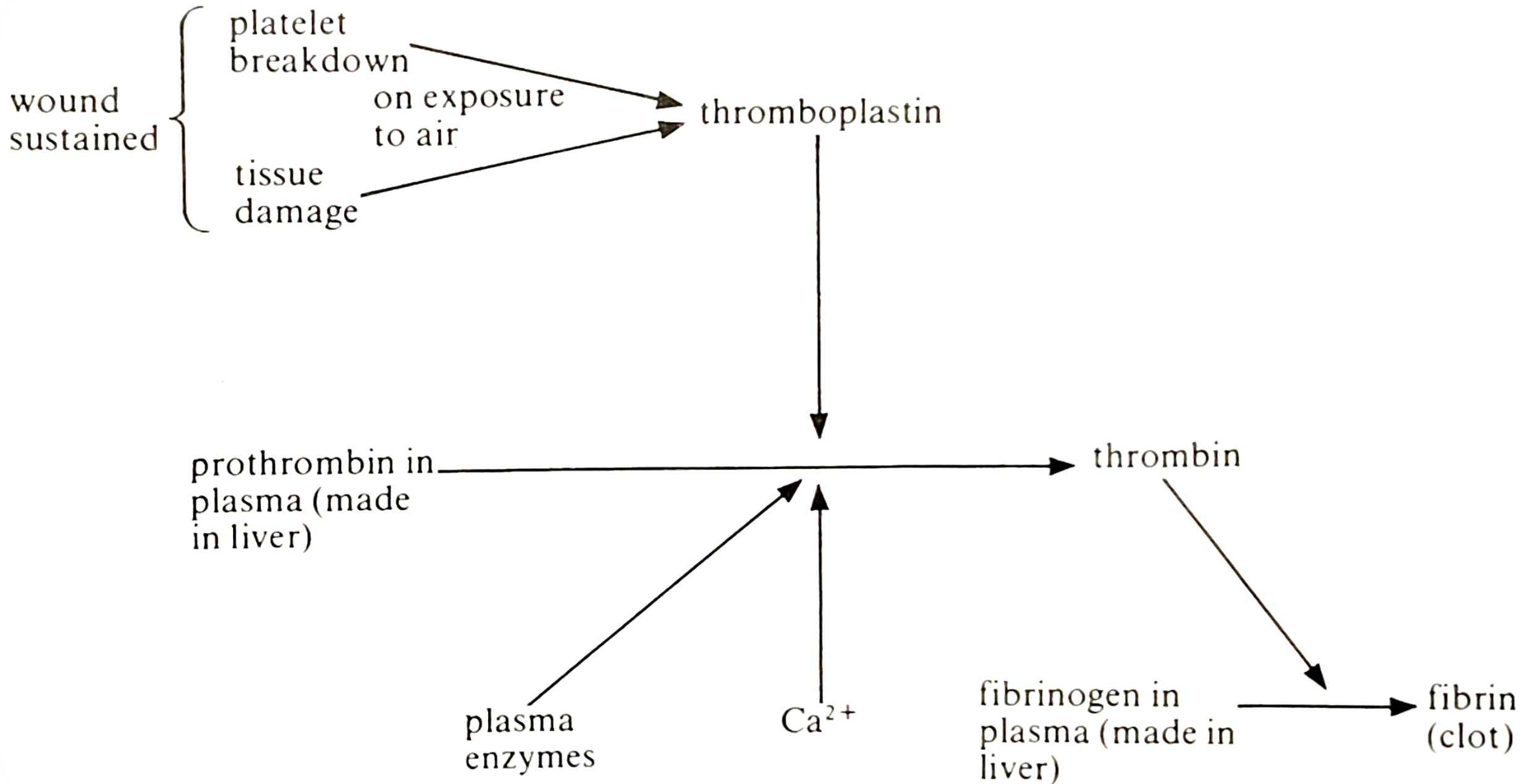
Blood escaping from a skin wound is exposed to air and mixes with substances oozing from the damaged cells and ruptured platelets.

These substances includes:

- ✓ Thromboplastin – a lipoprotein from injured tissues.
- ✓ Clotting factors VII & X – enzymes found in plasma.
- ✓ Calcium ions

Coagulation Cascade

- Thromboplastin, clotting factors and calcium ions, catalyze the conversion of prothrombin to **thrombin**.
- ❖ Prothrombin is a protein found in plasma, thrombin is its active form which is a protease.
- Thrombin hydrolyses fibrinogen (a large soluble protein) to **fibrin**.
- ❖ Fibrin is insoluble and fibrous in nature forms a meshwork which traps red blood cells.



Coagulation Cascade

- ❖ The meshwork made of fibrin traps RBCs and platelets, it later dries to form a scab.
- ❖ The scab prevents the entry of pathogens and allows the body to repair the damaged tissue.
- ❖ Therefore, absence or a low concentration of any of the essential clotting factors can lead to excessive bleeding. E.g. haemophilia.

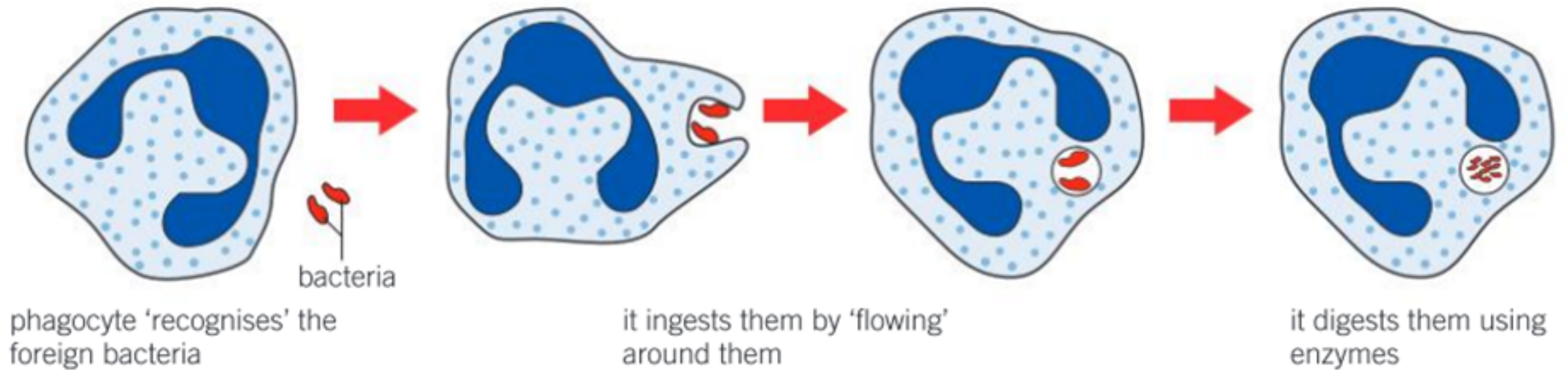
Phagocytes

When there is a breach in the skin barrier, phagocytes migrate from the blood to the site of injury.

Their role is to **engulf** “eat” and **digested pathogens** through the process of *phagocytosis*, killing the pathogenic microorganism.

Blood flow is increase to the site of infection and capillaries dilate, allowing more phagocytes to reach the area of infected tissue and makes the capillaries more permeable.

Phagocytosis



Immunity

Immunity

The blood plays an important role in protecting the body and is responsible for the immune response and immunity.

The substance that initiates an immune response is called the **antigen**.

The immune response involves lymphocytes producing **antibodies** towards the antigen.

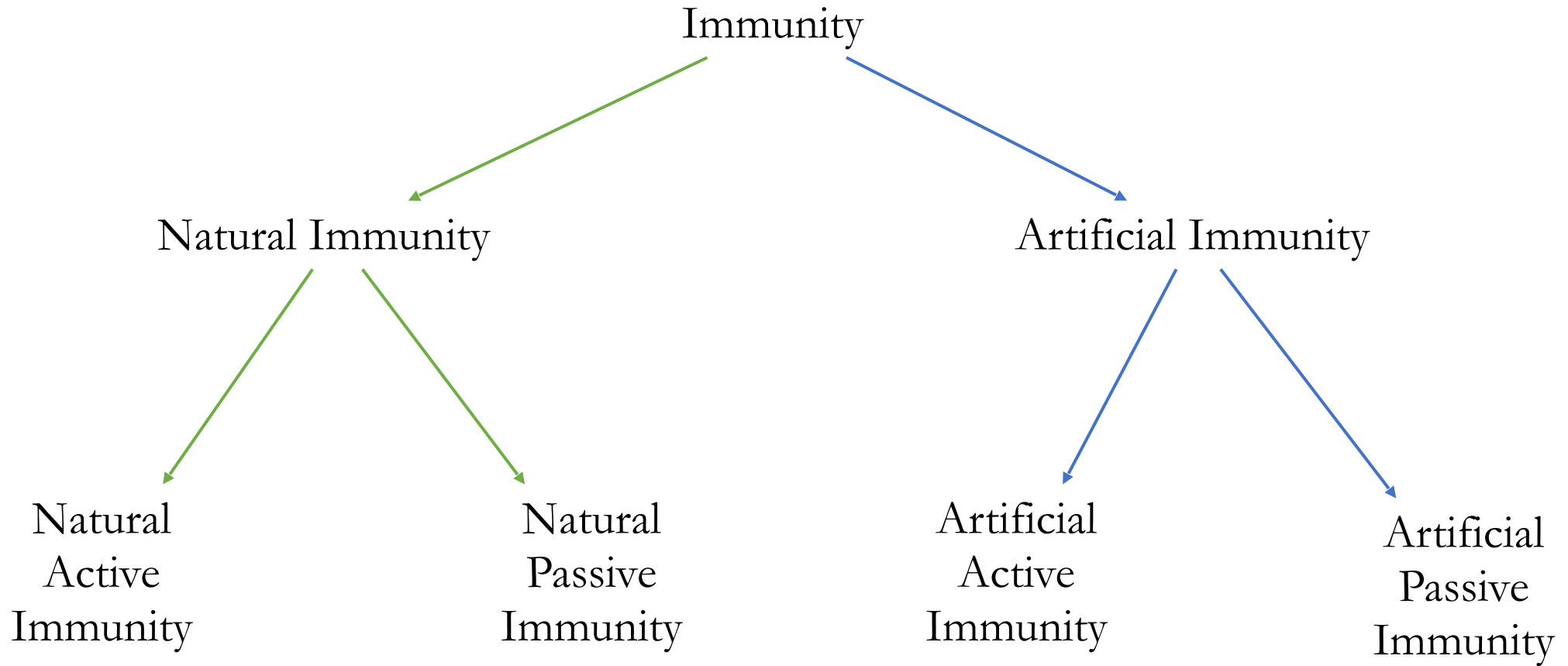
There are two types of immunity; natural & artificial.

Why is the Immune Response Important?

The immune response is specific where antibodies are produced to attack the antigen that has entered the body only. Therefore, defending the body against disease.

How does it occur?

- *Antibodies cause the antigens to clump together which can result in their immediate death or it slows them down just enough for the phagocytes to engulf and digest them.*
- *Antibodies can neutralize toxins produced by the antigens.*
- *Antibodies can prevent antigens from entering the body cells.*



What is Natural Immunity?

How does it happen?

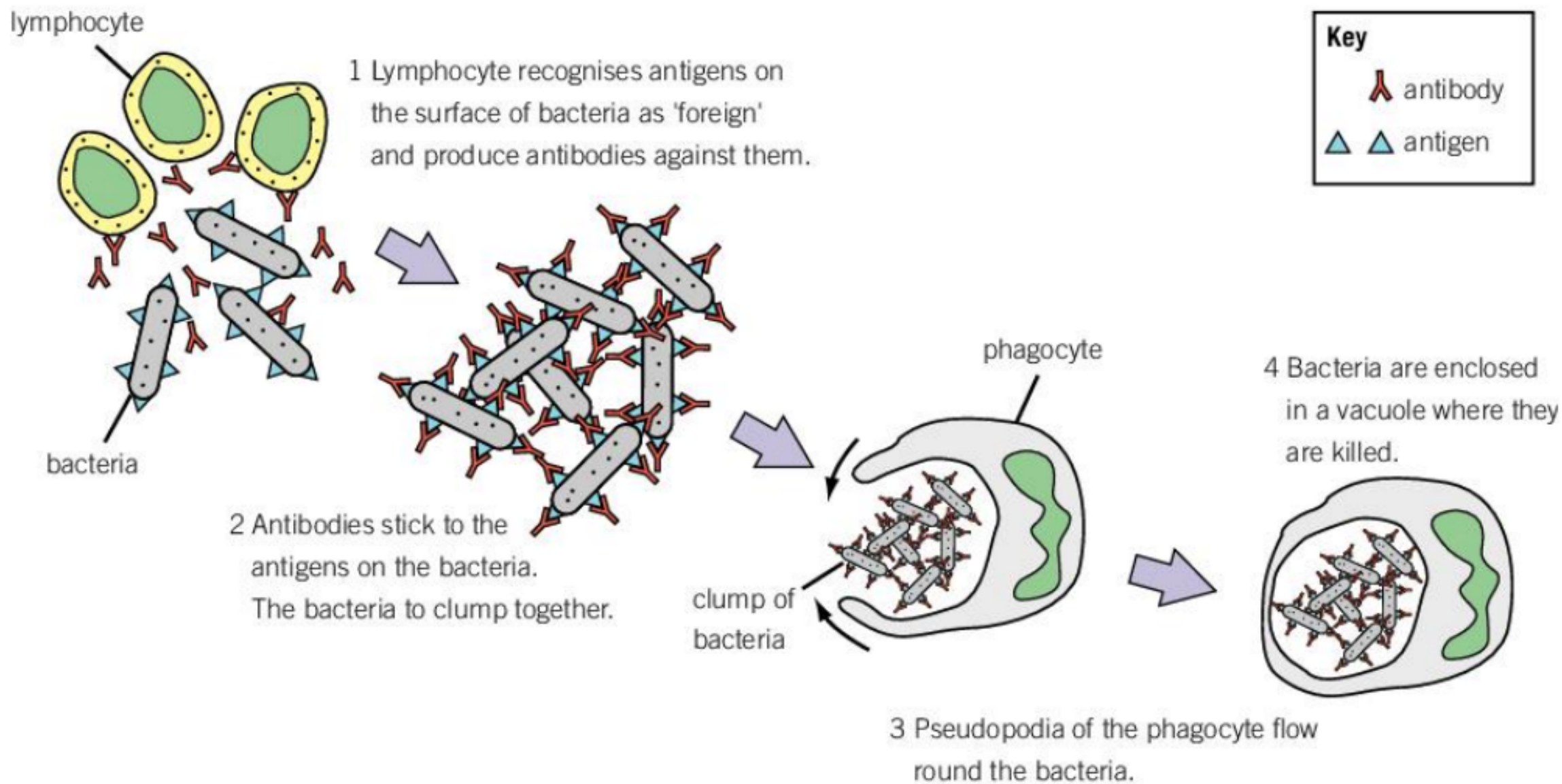
Natural Active Immunity

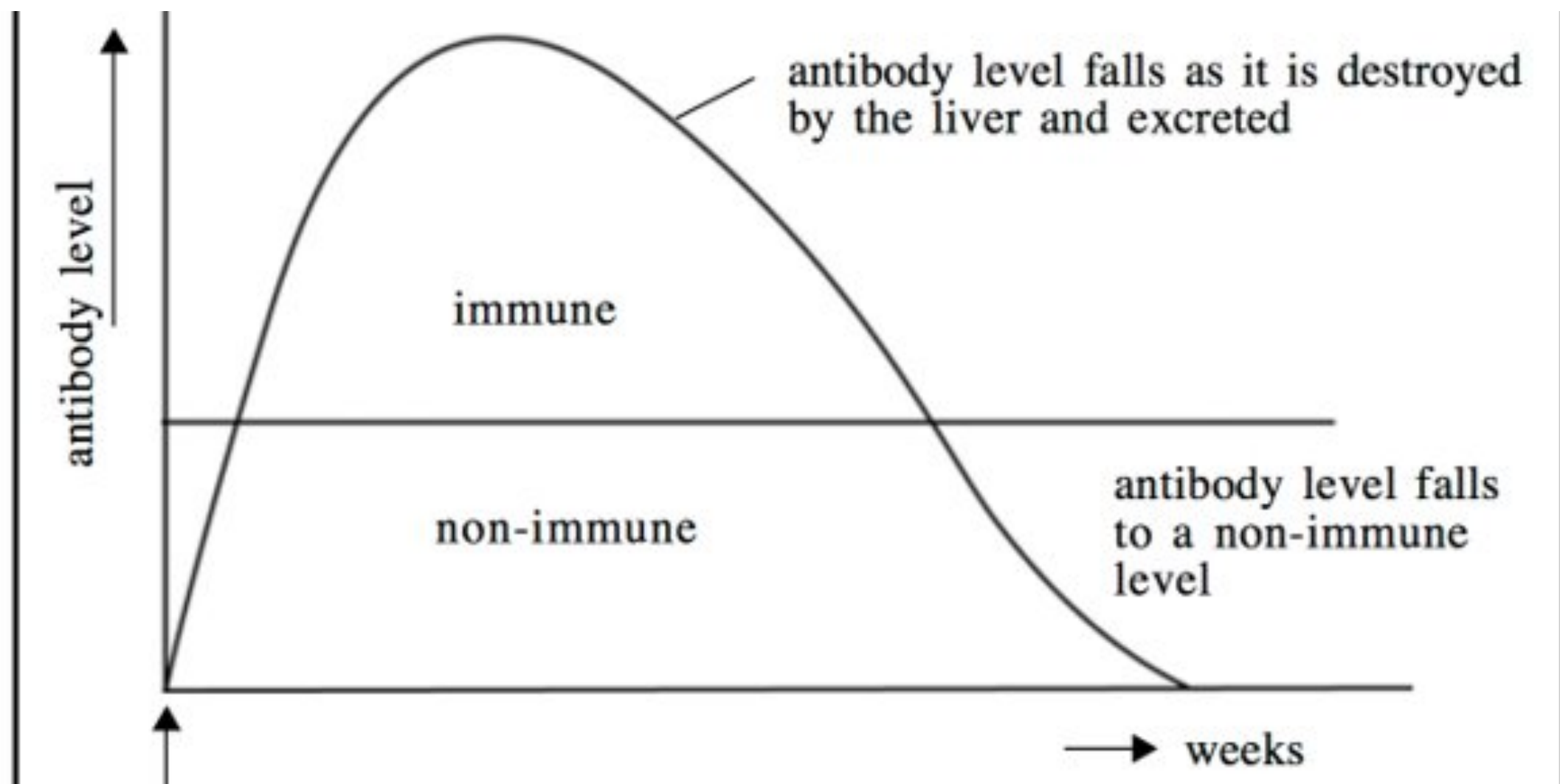
When a person becomes infected with a pathogenic disease, the lymphocytes takes time to recognize and understand the antigen before it can begin antibody production.

During this time, the individual will experience symptoms of the disease.

Once the lymphocytes have understood the antigen, they begin antibody production to neutralize and/or destroy the pathogen.

Some of the lymphocytes are converted into **memory-lymphocytes** which can recognize the antibody immediately and begin antibody production on next exposure.



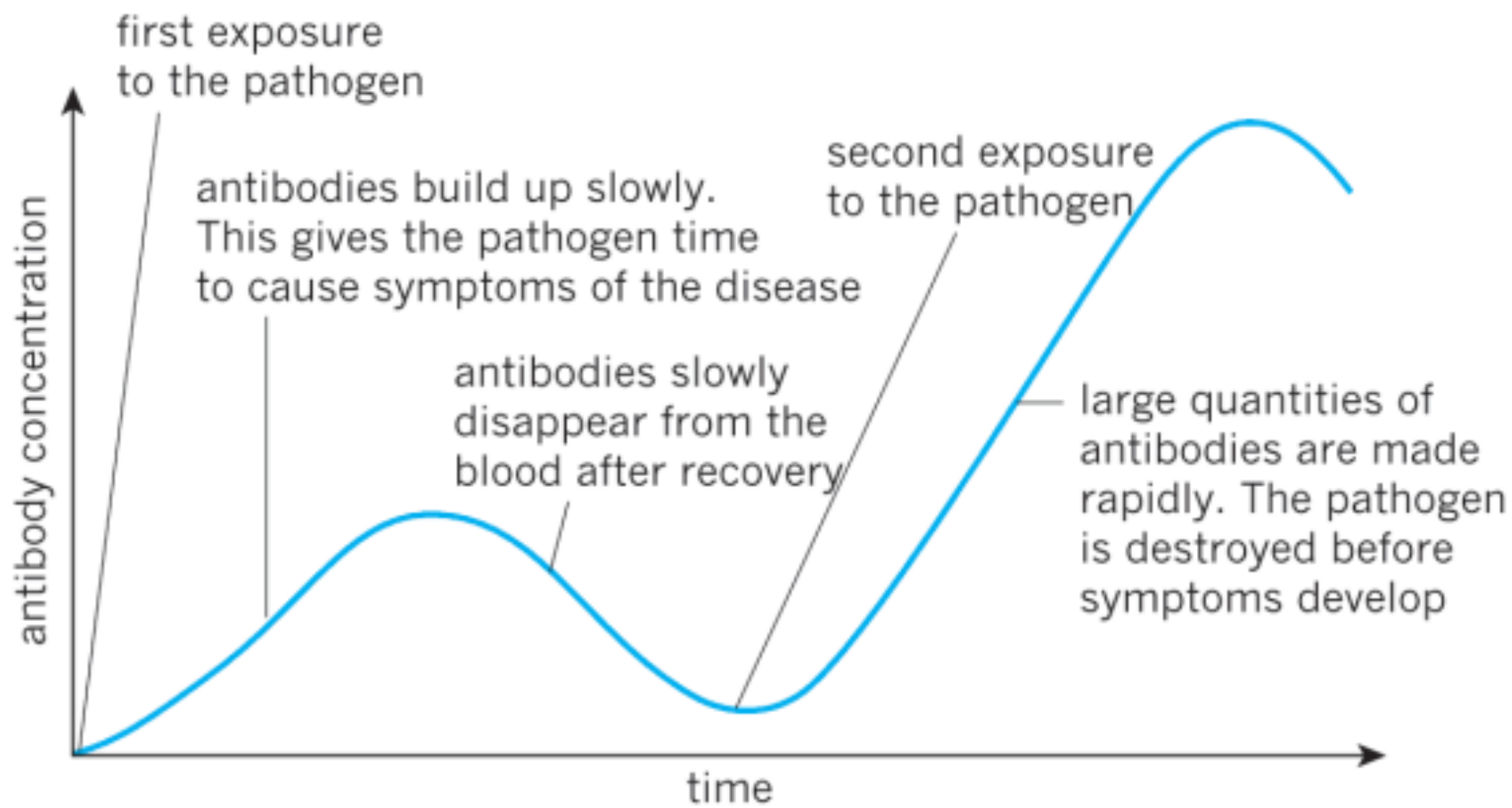


Natural Active Immunity

As time passes the antibodies as well as some of the lymphocytes will be destroyed by the liver.

However, for those that remain in circulation, will start producing antibodies upon reentry of the same antigen.

At this point, the individual is said to be resistant to that pathogen.



Natural Passive Immunity

Natural passive immunity occurs in babies where they receive antibodies from the mother.

- ❖ Antibodies from the mother passes to the fetus across the *placenta*.
- ❖ Antibodies also pass from the mother in the *breast milk*.

What do you think is Artificial Immunity?

Artificial Immunity

Artificial immunity is acquired by vaccination and is used to control the spread of communicable diseases.

A vaccine contains:

- * A dead pathogen/micro-organism.
- * Or a weakened form of the pathogen (live attenuated vaccine).
- * Together with the antigens from the pathogen.
- * It can also contain antibodies from another organism in response to infection by the pathogen.

Active Artificial Immunity

Vaccines work by injecting a *weakened or dead* form of the **pathogen** containing its **antigens**.

Therefore, it does not cause the disease but instead causes an immune response by the presence of antigens.

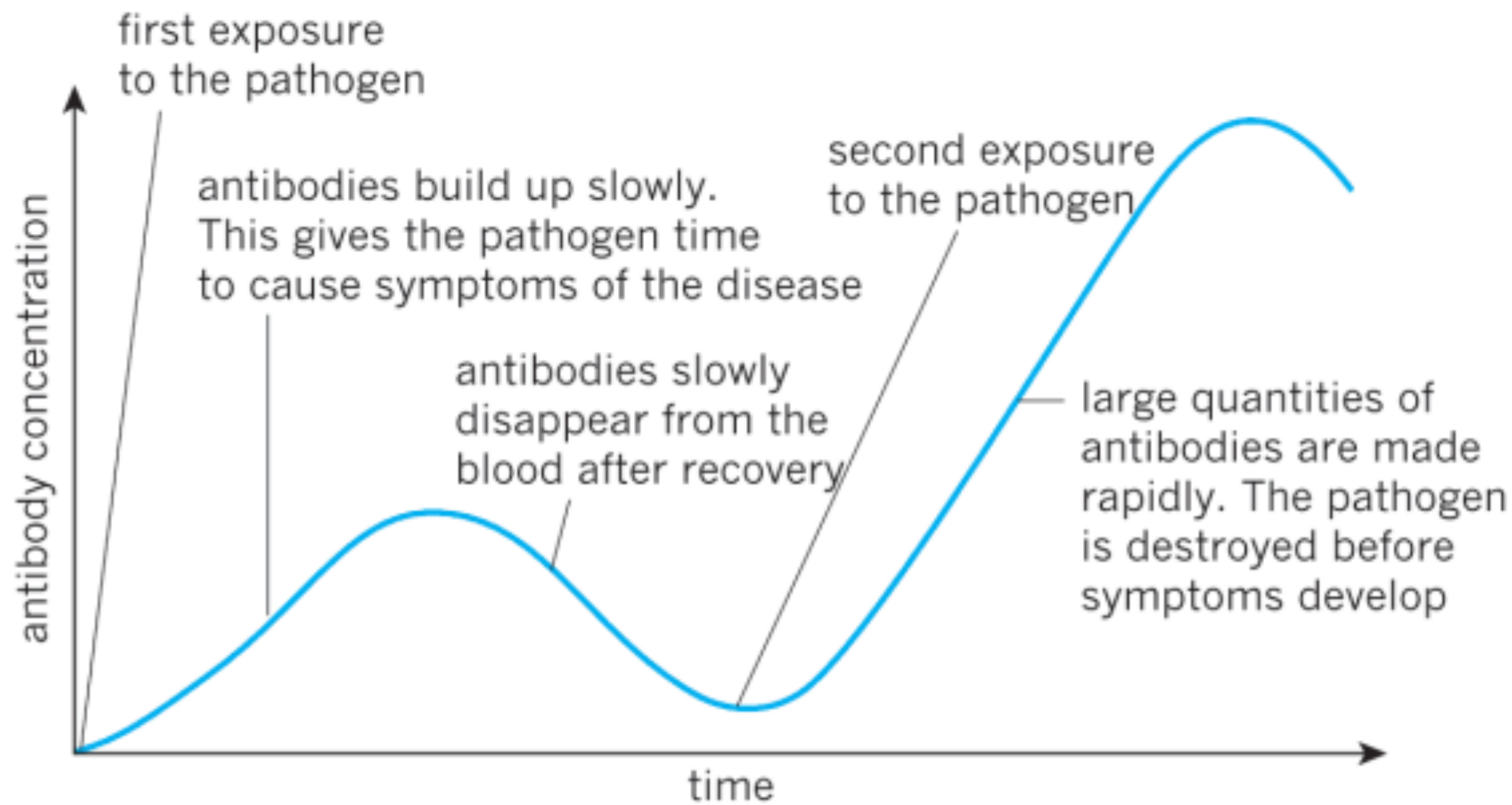
The lymphocytes begins producing antibodies after understanding the antigen and in the process produces **memory-lymphocytes** that are ready for when the pathogen enters the body.

Active Artificial Immunity

Likewise, after some time, the memory-lymphocytes will be destroyed in the liver after their lifespan.

This is why vaccines have boosters. It encourage the body's defense system to have memory-lymphocytes continuously circulating in the body.

Immunity can last from a short time to lifetime based on the vaccine (and its boosters).



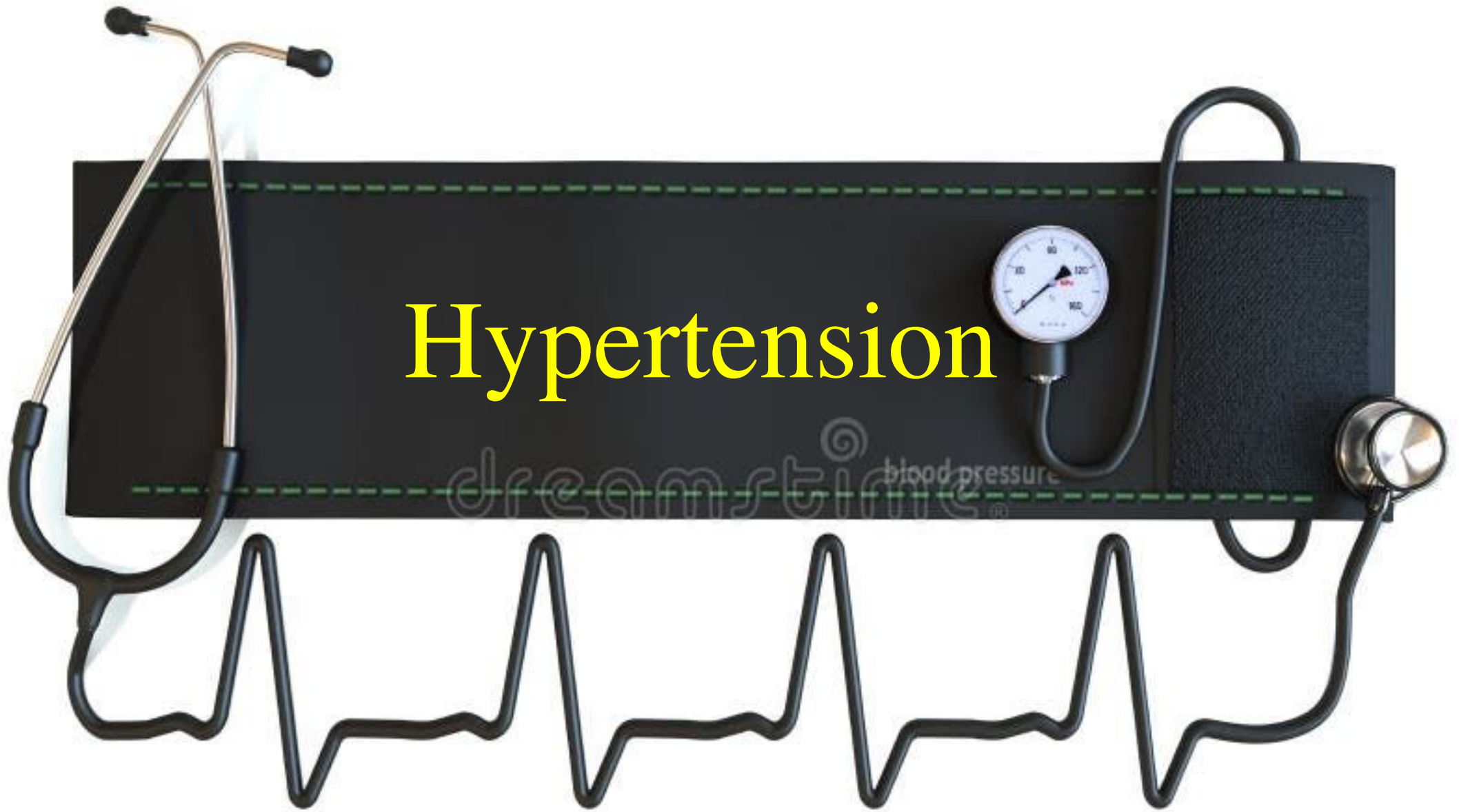
Artificial Passive Immunity

In passive artificial immunity, the vaccine contains the *antibodies/antitoxins* for the pathogen.

This is given in conditions where when the person encounters the pathogen, the pathogen kills the individual before the body can make antibodies to fight the pathogen.

Examples include tetanus toxin and leptospirosis.

Hypertension



Hypertension

- ♣Hypertension or high blood pressure is of a major concern in the Caribbean as it often causes death by heart attacks and strokes.
- ♣Hypertension can cause damaged to the smooth lining of the blood vessels resulting in hardening of the arteries. This causes the heart to pump harder and faster.
- ♣The effect of which can damage or even rupture blood vessels in the heart, brain and kidneys.

Hypertension

A stroke can occur with bleeding in the brain leading one-sided paralysis of the body, loss of speech and even death.

Hardening of the kidney arteries reduces the amount the fluid that can be filtered leading to the accumulation of waste products in body causing disease and even death or kidney failure.

If a coronary artery in the heart is blocked by plaque, reduced blood flow to the heart leads to permanent damage of the heart muscle. This is referred to as a heart attack.

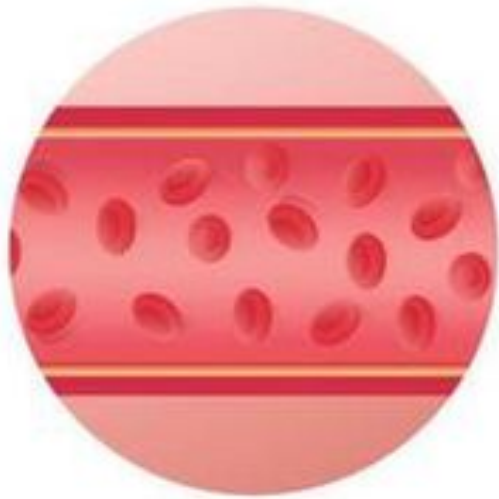
Hypertension & Atherosclerosis

Atherosclerosis or the *hardening of arteries* occurs when the lumen of the artery becomes narrowed by plaque and fatty deposits like cholesterol.

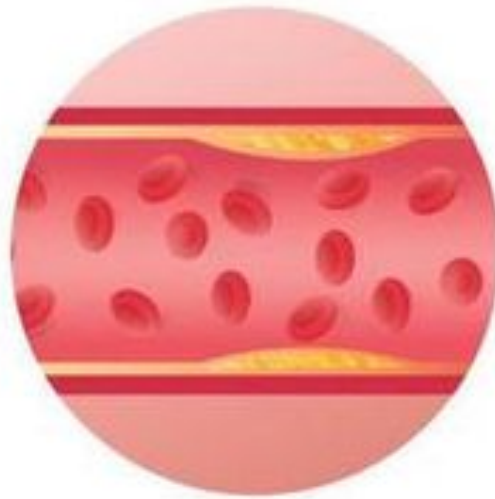
A narrowed lumen put strain on the heart muscle as it now needs to pump harder and faster to get the blood to the organs.

In doing so, this can cause artery vessel rupture as the elasticity of the artery is greatly reduced as well as stroke or heart attack as pieces of plaque can break off and get stuck in these smaller blood vessels.

STAGES OF ATHEROSCLEROSIS



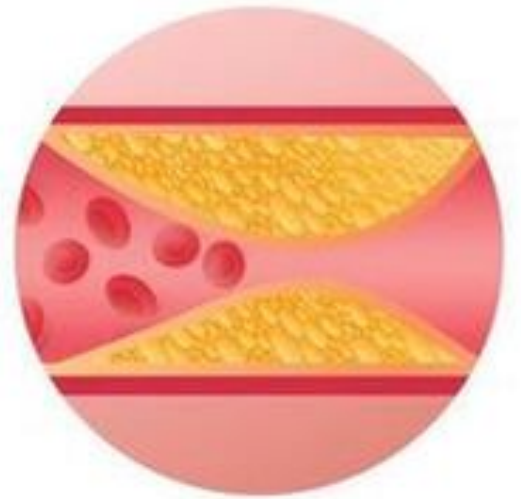
Healthy



Fatty Streak



Fibrofatty Plaque



Complicated Plaques

Factors which predisposes someone to Hypertension

✓ Stress

✓ Genes

✓ Diet

✓ Obesity

✓ Smoking